

Review article

Secure communication in IOT-based UAV networks: A systematic survey

Jatin Sharma, Pawan Singh Mehra *

Department of Computer Science and Engineering, Delhi Technological University, New Delhi, India

ARTICLE INFO

Keywords:

Internet of Things (IoT)
Unmanned aerial vehicle (UAV)
Flying adhoc networks (FANET)
Drones
Ground control station (GCS)
Secure communication
Survey

ABSTRACT

IoT (Internet of Things) has revolutionized the world with its applications like home automation, transportation, agriculture, etc. IoT-based Unmanned Aerial Vehicle (UAV) networks are an emerging field that introduces the UAV network with the power of the Internet. IoT-based UAV networks are a network of interconnected UAVs which are then equipped with sensors, a microcontroller to exchange collected data with each other, and a GCS (Ground Control Station) over the internet. IoT-based UAV network system is used for surveillance, monitoring, payload delivery, and much more which generate a plethora of sensitive information which can be obtained by adversaries in the middle. The communication between UAVs and GCS is vulnerable to security threats using jamming and eavesdropping attacks. The message between UAV and GCS is not enciphered which makes them error-prone. Attacks like GPS (Global Positioning System) spoofing can be possible by sending fake coordinates to trick UAV's control. Although much effort has been made toward UAV security. However, there is a dearth of descriptive reviews on secure communication in IoT-based UAV networks. The purpose of this paper is to find and examine peer-reviewed literature that discusses previous six-year established papers with existing attacks such as physical, and logical attacks with suggested solutions such as trajectory planning, lightweight schemes, solutions based on blockchain, quantum cryptography, etc. This paper analyzes the secure communication network between UAVs by systematically answering research questions based on the research methodology for the relevant study. Finally, the paper addresses several issues and guidelines for future research.

1. Introduction

Drones, also known as unmanned aerial vehicles or UAVs, were initially created for military use. A drone is a flying object that has remote control independence and self-piloting capabilities. The drones are dynamic and fast, so maintaining communication reliability is essential which can be achieved through the Mavlink protocol by bridging the gap between UAV and GCS (Ground Control Station). IoT explains a network of interrelated things that is equipped or embedded with sensors, microcontrollers, and software to share data with other physical devices over the internet. Therefore, IoT-based UAV networks have similar properties to IoT. IoT-based UAV networks are multi-UAV systems which are a group of interconnected UAV nodes with the power of the internet and equipped with embedded sensors and microcontrollers. As per the Federal Aviation Administration, there were about

* Corresponding author.

E-mail address: pawansinghmehra@gmail.com (P.S. Mehra).

<https://doi.org/10.1016/j.iot.2023.100883>

Received 28 April 2023; Received in revised form 27 June 2023; Accepted 17 July 2023

Available online 22 July 2023

2542-6605/© 2023 Elsevier B.V. All rights reserved.

855,860 UAVs registered in the United States. The annual growth of the drone market is estimated to 6.4% approximately which will be expected to be double by 2024. The usage of IoT-based UAV technology in the supply chain, heavy payload delivery, photography, rescue, inspection services, and military operations have gained huge attention, which is hard to accomplish with a single UAV. So these UAVs need to transfer commands with one another for maintaining coordination. IoT-based UAV networks generate a large amount of data including telemetry, imagery, and sensor readings. Therefore, Implementing numerous steps to safeguard the confidentiality, integrity, and availability of data exchanged between UAVs and ground control stations lays the foundation for secure communication in IoT-based UAV networks. Existing studies [1–5] of UAV security, privacy, and communication designs have mostly focused on cryptographic techniques which are computationally expensive strategies. We have also found that major risks associated with IoT-based UAV Networks such as physical and logical attacks like net throw attacks, Sybil, Sink holes attacks, and cutting-edge solutions like Lightweight mutual authentication, blockchain, and quantum cryptography for their privacy and security have received very little attention. The prior work's primary drawback is the absence of a thorough investigation of the assault's duration cycle and security weaknesses of UAV communication. To get around the shortcomings of the current security solutions, existing countermeasures must be examined, and fresh approaches must be suggested. So to fulfill the need for secure communication in IoT-based UAV Networks operations, this paperwork discusses the physical and logical attacks along with their prevention schemes such as trajectory planning techniques, lightweight authentication, strong encryption methods, etc. with future directions. Based on existing work limitations, the research questions are formulated to give the researchers a clear point of view to understand the secure communication scenario through this systematic survey.

1.1. Motivation

Despite the obvious advantages of UAV networks, their implementation is constrained by security concerns. In 2011, Unmanned aerial vehicle security was raised when Iranian troops captured an American RQ-170 [6]. The risk of enemies initiating network assaults exists whenever UAV networks are used for tasks like combat communication and scouting in dangerous circumstances. During the Russia–Ukraine war in 2022 [7], the army of Ukraine formed small teams to physically attacks the Russian drones with the help of laser weapons. The author [8] explains how attackers can take advantage of tampering with the communication between UAVs and their remote counterparts, rendering them vulnerable. With recent perusal, it is clear that IoT-based UAV Communication is not secured. UAVs with sensitive information on military missions can be compromised. As far as security is concerned, high-end missions like military applications viz. reconnaissance, search, rescue, and power line inspection require more safety, and existing studies provide us with brief assessments or surveys with restricted to only a few security attacks or countermeasures. Since no systematic study in the field of IoT-based UAV Networks was released, so this is a recent study. By offering a systematic survey for security solutions about many security challenges, we want to close the gap. This motivated us to carry out this study to fulfill the gaps and security needs in the high-end application of UAVs. According to the identified gaps in the literature review, the research questions (RQ1–RQ5) in Section 1.2.2 are formulated based on security breaches in real-life scenario UAV applications due to which there is a need for preliminary study of recent attacks and the communication technologies which keeps the power-constrained UAVs operable and comparative analysis are required to make familiar the researchers about state of art technologies to mitigate these attacks.

1.2. Research aim and research questions

This segment furnishes details regarding the research aim & queries through a thorough review of Unmanned Aerial Systems. The Research aims and questions are defined as:

1.2.1. Research aim

To study the security vulnerabilities and solutions in communication in IOT-based UAV networks to raise awareness among researchers and the general community.

1.2.2. Research questions

Research Questions to meet our objectives are as follows:

- **RQ1** *What are the various security vulnerabilities or attacks through which communication in IoT-based UAV networks is compromised?* To answer this question, the most recent research articles are analyzed in Section 2 to figure out various security breaches in UAV networks.
- **RQ2** *What is the cutting-edge status in terms of high-end UAV security?* The preliminary study and analysis of recent UAV attacks in Section 2.3 responds to the query. Consequently, it is necessary to carefully observe the recent assaults, and research findings and assess their relevance to this study.
- **RQ3** *What communication technologies are employed to operate UAVs?* To answer this question, the different types of control channels like Bluetooth, WIFI, etc, have been studied and Section 3.1.2 will address this query.
- **RQ4** *What makes this survey different from the existing survey?* To answer this question, the analogy is mentioned with the existing study based on parameters defined in the tabular layout, and Section 4 will address this query.
- **RQ5** *What are the current defense strategies available for ensuring secure communication in IoT-based UAV networks?* The research question is answered based on previous research question one (**RQ1**) about security vulnerabilities. Section 6 discusses the related work based on recent defense strategies and provides a comparison table with existing research.

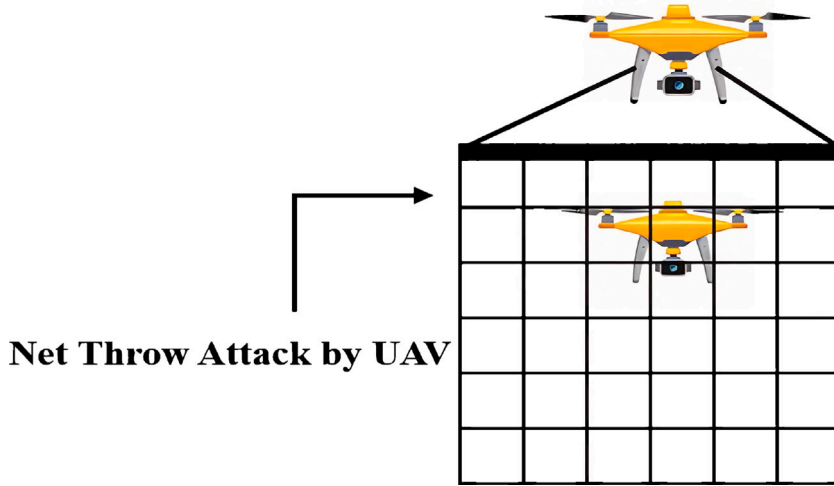


Fig. 1. Physical attack.

1.3. Outline of the paper

The sections are arranged in the following manner: Section 2 gives an outline of the pertinent investigations on security vulnerabilities. Section 3 discusses the overview of UAVs and FANET. Section 4 presents the related work and their comparison with this survey. The research methodology is presented in Section 5. The state of art countermeasures for secure communication in IoT-based UAV networks is presented in Section 6. Current research issues and future directions are presented in Section 7. Section 8 presents the concluding remarks.

2. Preliminary study of attacks on UAV

This section discusses the related studies about security vulnerabilities and explains query RQ1. Multiple security vulnerabilities can exploit UAV communications. Two kinds of attacks are possible as follows.

2.1. Physical attacks

The control of a UAV can be taken by another drone called an interceptor drone. A UAV's task could be jeopardized if it collides with moving or immovable objects, as well as with other aircraft. An idea proposed is to equip the second unmanned aerial vehicle with a spear, which could be used to aim at the intrusive UAV. Alternatively, a mesh could be thrown over it [9], or a wire could be released into its propellers [10] to pull it away and eliminate the threat. Other types of physical attacks are directed energy weapons like lasers and microwaves [7]. The physical attack demonstration is shown in Fig. 1.

2.2. Logical attacks

As UAVs are unmanned vehicles so they are operated using commands forward by the GCS (Ground Control Station) to control flight speed and altitude etc. The various type of logical attacks is discussed below:

- **GPS Jamming:** A jamming attempt stops a receiver from picking up the real GPS signals. This can be done by transmitting higher-powered interference signals in the same frequency band [6].
- **Meaconing:** Meaconing is the practice of recording real GPS signals and relaying them to the receiver with an additional delay [6].
- **Spoofing Attack:** In this kind of attack, the attacker sends a signal that is malicious and more powerful than the real signal, tricking the receiver into using the fake signal. It is a more destructive attack as UAV control can be taken by adversaries. Interestingly, US RQ 170 Sentinel was captured by Iranian forces after a recent security attack against UAVs [6].
- **Eavesdropping:** The adversary can directly access the exchanged UAV data in the open environment due to the absence of encryption and other security measures [11].
- **Code injection attack:** It is an active attack in which the adversary injects fake information and instructions by masquerading as a legitimate entity [12].
- **Virus:** A virus is a passive attack and self-replicate program which spread from computer to computer. In [13], a virus known as a keylogger has infected a group of U.S. military drones stationed at an Air Force base in Nevada, enabling it to monitor every key and button pressed by their pilots.

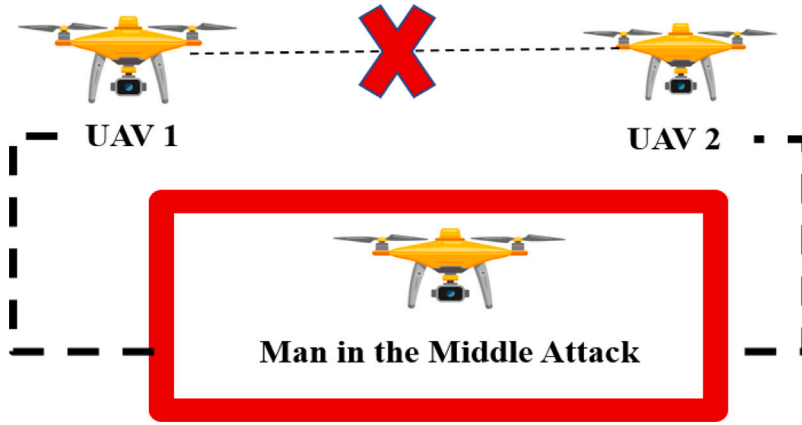


Fig. 2. Logical attack.

Table 1

Type of attacks.

Sno.	Type of attacks discussed	[6]	[16]	[17]	[15]	[18]	[8]	[11]	[12]	This survey
1	GPS Jamming	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
2	GPS Spoofing	Yes	Yes	No	No	No	Yes	Yes	Yes	Yes
3	Meaconing	Yes	No	No	No	No	No	No	No	Yes
4	Eavesdropping	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
5	Injection Attack	Yes	No	No	Yes	Yes	Yes	Yes	Yes	Yes
6	Virus	No	No	No	No	No	No	No	Yes	Yes
7	Active Eavesdropping	No	No	No	No	No	Yes	Yes	Yes	Yes
8	Replay Attack	Yes	No	Yes	No	No	No	Yes	Yes	Yes
9	De-authentication attack	Yes	No	No	No	Yes	Yes	Yes	Yes	Yes
10	Black Hole	No	No	No	Yes	Yes	No	Yes	Yes	Yes
11	Grey Hole	No	No	No	Yes	No	No	Yes	No	Yes
12	Worm Hole	No	No	No	No	No	No	No	Yes	Yes
13	Sink Hole	No	No	No	Yes	No	No	No	No	Yes
14	Sybil Attack	No	No	No	No	No	No	Yes	Yes	Yes

- **Man in the Middle attack:** This is a type of active attack in which the communication link between the multi-UAV system is intervened and a malicious drone acquires the control by modifying and dropping the content [11].
- **Replay attack:** The malicious UAV can pose as a valid sender after a surveillance attempt and transfer the encrypted data to another UAV. In [14], the replay attack was performed on the E010 drone by acquiring four channels with 20 MHz bandwidth radio signal and by replaying it through HackRF hardware.
- **De-authentication attack:** In this kind of attack, the adversary sends the de-authentication frames to interrupt the target UAV connection. In [6] the Aircracking was used to de-authenticate or disconnect a genuine client and gain control over the system.
- **Denial of Services (DoS) attack:** In this type of attack, the adversary makes the UAV network unavailable. The adversary obstructs the UAV network by flooding it with data [6]. The types of DoS attacks are the following:
 - **Black Hole Attack:** In this type of denial-of-service attack obtains vicious UAV packets but does not transfer them to the target UAV and discards them [15].
 - **Grey Hole Attack:** It is the type of DoS attack that obtains vicious UAV packets and chooses the packets and transfers the chosen packets to the target UAV or neighboring UAV and obliterates the rest of the packets [15].

Some other types of attacks:

- **Wormhole attack** In this type of attack, the adversary makes a tunnel with the help of one or more malicious nodes and then they use tunnels to forward packets to other networks [11].
- **Sinkhole attack** In this kind of attack the adversary tries to gather network traffic by broadcasting on the fake path with altering routing information [15].
- **Sybil attack:** When a malevolent unmanned aerial vehicle infiltrates an existing network, it generates a digital resemblance. such I1,I2,I3 [11].

The logical attack demonstration is shown in Fig. 2. The hierarchy of attacks is shown in Fig. 6 and a related study of attacks is shown in Table 1 with Yes and No values Yes means “Attack is discussed in respective work” and No means “Attack is not discussed”.

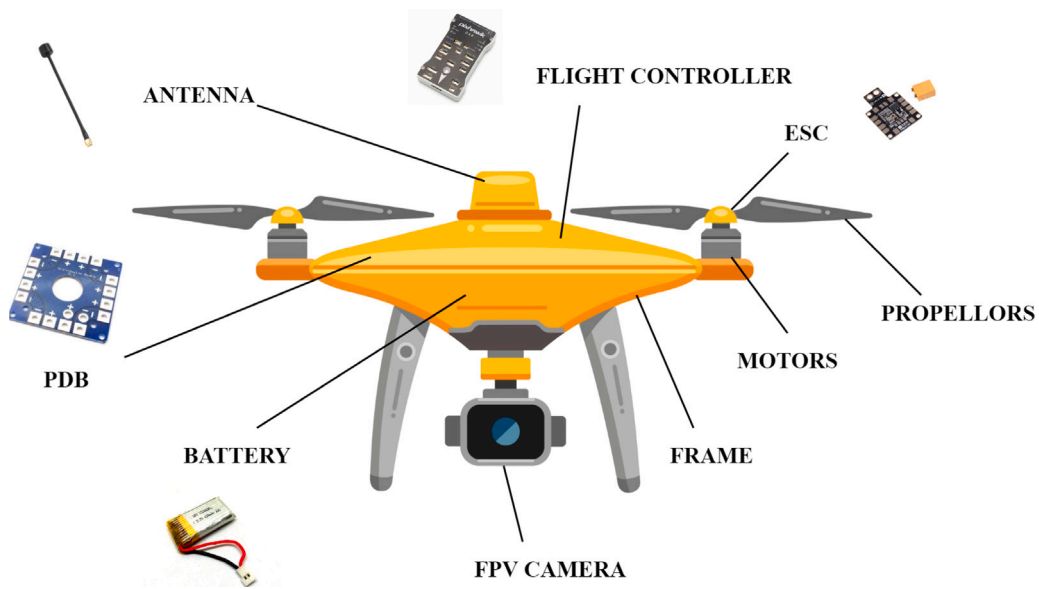


Fig. 3. UAV.

2.3. Recent drone attacks

This section elucidates the current state of the art regarding high security and addresses Query RQ2.

- **SkyJack:** SkyJack takes advantage of an open WiFi communication channel that exists between the Parrot AR drone 2, which controls the UAV, and the smart phone. An attacker may introduce de-authentication packets into the system and cut off the authorized operator from the UAV [19].
- **ProtoX:** A ProtoX is a small quadcopter drone that could be reversed engineered with equipment like a logical analyzer and development board. To control the small quadcopter, the adversary removes the microcontroller of the remote control with joysticks and connects it to the output pins of a development board. This allows them to write program scripts to transmit control signals to the quadcopter. Thus, reverse engineering can be possible [19].
- **MalDrone:** A drone can be compromised if malware is installed on the device in case the communication channel is not encrypted. One restriction is that the malware can be installed on Parrot AR.2 Drone. The malware can attack the flight controller and it allows the attacker to connect back to control [19].
- **War Flying:** By attaching wireless communication devices to the UAVs which allows the adversary to spy on objects or targets. Bluetooth, WiFi, and GSM (IMSI Catcher) are used for privacy invasion [20].
- **WiFi and Bluetooth Cracking:** In 1997, WiFi was introduced, and it utilized a cryptographic protocol known as WEP (Wired Equivalent Privacy). However, this had certain limitations in terms of security and was susceptible to attacks. To address these issues, a new encryption standard called WiFi Protected Access 2 (WPA2) was developed, which provides a high level of security. Nonetheless, depending on certain conditions, this standard can still be compromised [6]. Bluetooth technology is often used for exchanging data between aircraft, micro UAVs, and remote controls. However, this technology can be susceptible to various security attacks [21], including the newer Bluetooth Low-Energy (BLE) standard.

3. Overview of UAVs and FANET

This section covers all the background information about unmanned aerial vehicles and FANET which answers RQ3.

3.1. UAV

UAV is an unmanned aircraft or vehicle which can be operated autonomously as well as independently. UAVs are famous for their role in reconnaissance and rescue and battlefield. The UAV's network covers various domains like wireless hotspot services, smart cities, remote sensing, agriculture, etc. The UAV components are presented in Fig. 3:

The main components of UAVs are:

- Frame
- Motors
- Electronic Speed Controllers (ESC)



Fig. 4. Aircraft based on wings.

- Flight Controller (FC)
- Power Distribution Board (PDB)
- Radio Receiver
- Battery
- Propellers
- FPV Camera
- Video Transmitter
- Antenna

3.1.1. Types of UAVs

UAVs are categorized based on the functional area of wings, weight, altitude and range:

According to Wings [5]:

- **Fixed wing:** This is a small version of aircraft without a pilot having fixed wings that need a runway to fly and land. They have long range and endurance.
- **Rotatory wing:** The rotatory wing craft can be single as well as multirotor or VTOL UAVs. This type of wing has high maneuverability. It is easy to take off with precise control.
- **Hybrid wing:** It is the combination of both fixed as well as rotatory wings.

The category of UAVs based on wings is shown in Fig. 4:

According to Weight [22]:

- **Nano:** aircraft with a *weight* ≤ 250 g
- **Micro:** aircraft with a *weight* > 250 g and ≤ 2 kg
- **Small:** aircraft with a *weight* > 2 kg and ≤ 25 kg
- **Medium:** aircraft with a *weight* > 25 kg and ≤ 150 kg
- **Large:** aircraft with a *weight* > 150 kg

According to Altitude [5]:

- **High altitude platform (HAP):** This type of aircraft operates at a very high altitude greater than 10 km. Ballons, aircraft, and airships come under this category.
- **Low altitude platform (LAP):** This type of aircraft is dynamic and flexible to fly at a height of up to 10 km. VTOL aircraft belong to this category.

According to Range[5]:

- **Close Range:** aircraft with a range of up to 10 km.
- **NATO:** aircraft with a range of up to 50 km.
- **Tactical:** aircraft with a range of up to 50 km.
- **Medium Altitude Long Endurance (MALE):** aircraft with a range of up to 200 km.

- **High Altitude Long Endurance (HALE):** aircraft with unlimited range.
- **Hypersonic:** aircraft with a range of more than 200 km.

3.1.2. Control channels

From a technical standpoint, there is a wide range of communication technologies that can be employed to regulate a UAV. The essential requirement is transmitting data from the RC to the UAV and vice versa, which can be accomplished through various types of electromagnetic radiation.

- **Infrared:** In households, Infrared (IR) technology is commonly used for TV and peripheral device remote controls, mainly because it is simple and inexpensive. However, its use for controlling Unmanned Aerial Vehicles (UAVs) is greatly limited due to the technology's physical constraints. The transmitter (light source) and the receiver (light detector) must have a line of sight, and any obstruction between them can cause reception issues. Utilizing Infrared light-based remote controls for UAVs can be difficult, given that they are usually operated at distances exceeding 10 meters from the remote control, and it is not always possible to avoid obstacles that may come between the remote control and the UAV. Additionally, IR communication can be affected by environmental factors such as rain and sunlight, further limiting its practical use in a typical outdoor Unmanned Aircraft System (UAS) setting.
- **Radiowaves:** Radiowaves are a type of electromagnetic radiation that have a longer wavelength than Infrared light. They are classified into various frequency bands based on their wavelength characteristics, and their usage is regulated by the National Telecommunications and Information Administration in the United States and the European Committee of the Regions in Europe. Table [23] displays the IEEE-defined frequency bands and Specific frequency bands which are prohibited for use due to regulatory restrictions, and certain frequencies are assigned solely to private entities through licensing. However, there are unlicensed frequency ranges that can be utilized to control Unmanned Aerial Vehicles (UAVs). The authors have provided an overview of the various frequency bands utilized in the European Union and their applications.
- **Bluetooth and WiFi:** UAVs are commonly operated through standardized technologies like WiFi or Bluetooth that function on unregulated frequencies. These technologies are typically employed to link a personal computer with drones to program the UAV's flight computer in advance. These technologies have a limited range due to their wavelength, making them suitable for controlling UAVs with restricted ranges, particularly Micro/Mini UAVs. The cost-effective method is advantageous because it allows customers to use their smartphones remotely, eliminating the need for additional RC.

3.2. FANET

FANET is the subset of MANET where the UAV nodes are connecting in an ad-hoc manner. Every UAV operates at a higher speed in comparison with MANET or VANET and Underwater ad-hoc networks. Each UAV consists of physical devices such as sensors, GPS, and a flight controller. FANETs is utilized in defense organization or military and civilian applications because of their dynamic nature, self-operatable, self-configured, etc. Fig. 5 presents the FANET architecture.

3.2.1. Communication in FANET

In a FANET, UAVs can communicate by exchanging data among themselves without any infrastructure and can be coordinated with Ground Control Station. There are four types of communication [24].

- **A2 A (Air to Air Communication):** In A2 A communication, one UAV can exchange data with another UAV and act as a relay node to extend the connectivity range.
- **A2G (Air to Ground Communication):** In this type of communication, one UAV can exchange essential data with the ground control station. The Ground control station can send control commands to the UAVs to perform a specific task.
- **S2 A (Space/Satellite to Air Communication):** Space-to-Air (S2 A) communication is the exchange of data between a UAV and satellites or space-based technologies. This sort of communication may be used in a variety of UAV tasks, including navigation, control, data transfer, and remote sensing.
- **Hybrid communication:** This kind of communication is the integration of the above three communications i.e. S2 A, A2 A, and A2G which is required for higher coverage and data rate.

3.2.2. Features of FANET

There are many features of FANET which are the following [24]:

- **Cost:** The cost of small UAVs in FANET is low because of their lower operating expenditure.
- **Survivability:** Multi UAVs in FANET is survivable in the case of one UAV node failure in a mission.
- **Speedup:** UAVs in FANET can accomplish the mission at a faster rate due to more in number.
- **Scalability:** The multi-UAVs network is scalable as more UAVs can participate in FANET and thus coverage area in the mission can be expanded.
- **Reliability:** Due to bad atmospheric conditions, the A2G communication is adversely affected and in that case, the A2 A condition can resolve the connectivity issues to ensure network reliability.

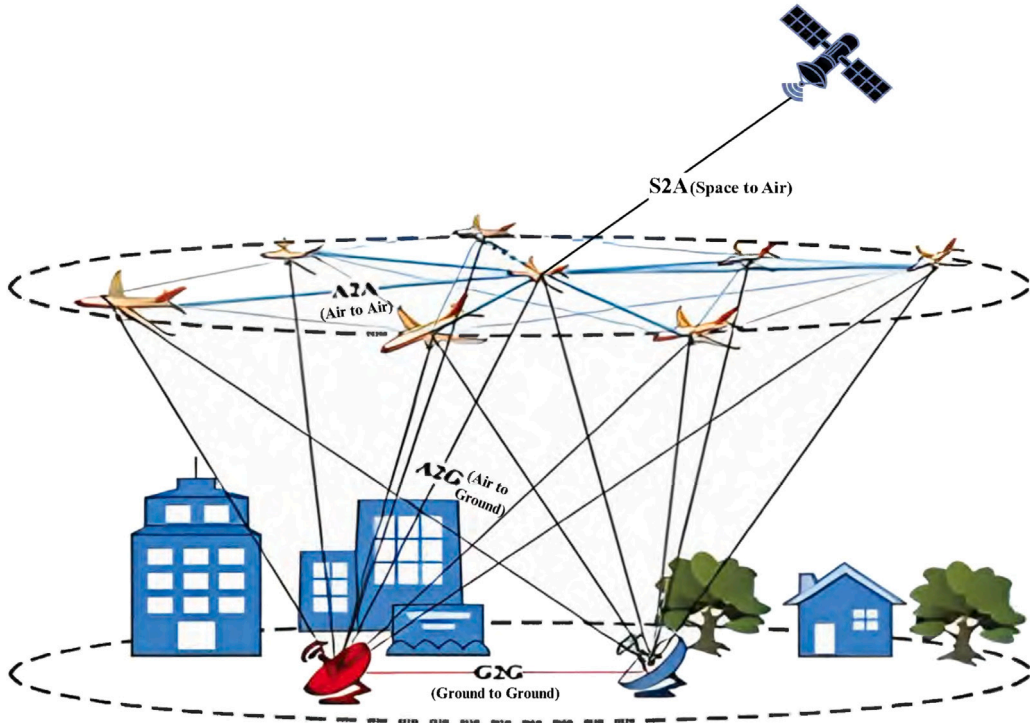


Fig. 5. FANET Architecture.

4. Related work

This section presents the related studies on secure communication in IoT-based networks. There are fewer publications in this field but this paper tries to cover the most relevant paper of the last six years with the comparison table as shown in Table 2 and address the query RQ4.

In [1] the author has discussed secure communication techniques i.e., hierarchical identity-based broadcast encryption and pseudonym technique but has not covered each attack and their countermeasures. Moreover, According to the abstract the author has mentioned that there is no usage of a third-party central server but the methodology has mentioned the usage of a third-party Remote Management Server (RMS). As per the computational assumption the author has implemented his scheme in bilinear groups which is hard to break but the cost of implementing the bilinear groups is high which is not feasible to implement. The Performance analysis is also performed in which less number of figures and tables were used for better understanding. The category of UAVs is also missing in this paper. Preliminary study of UAV is also missing in this paper. The proposed scheme in this paper has not gone through formal security analysis approaches like AVISPA, ProVerif, etc.

In [2] The authors have provided the cyber attacks countermeasures using machine learning algorithms like Neural networks but due to power-constrained UAV, the execution of complex operations is difficult. The authors have also spoken about cyber-attacks i.e. active and passive attacks such as modification and Denial of service attacks. Further, the author has described the detection schemes like rule-based specification and Bio-inspired detection schemes. The author has presented only one security frame architecture which is based on an intrusion detection system. The author has not described any lightweight encryption and cutting-edge countermeasures for IoT-based UAV networks. The author has not mentioned what type of security analysis has been done for the verification of the proposed framework.

In [3] The author has described the scenario of attacks on GPS (Global Positioning System) and ADS-B (Automatic Dependent Surveillance-Broadcast) systems(ground control station (GCS)) only without any focus on the security of sensors data. The author has not provided any privacy-preserving authentication scheme which is present in our paper. The logical attacks and their taxonomy like data interception, data manipulation, and DoS attacks but cannot discuss physical attacks [9]. The author has covered cryptographic-based defense methods like some lightweight cryptography methods but does not cover all kinds of the latest encryption techniques like ECC (Elliptic curve cryptography) which is a form of public-key cryptography that utilizes the mathematical properties of elliptic curves over finite fields to ensure security. ECC offers the advantage of using smaller keys compared to other forms of cryptography while still maintaining an equivalent level of security. Moreover, the author has not provided any efficient detection mechanism for the intrusion but our paper has provided a different intrusion detection mechanism. There are a lot of cutting-edge technologies which need to be mentioned by the author and which are covered in this paper.

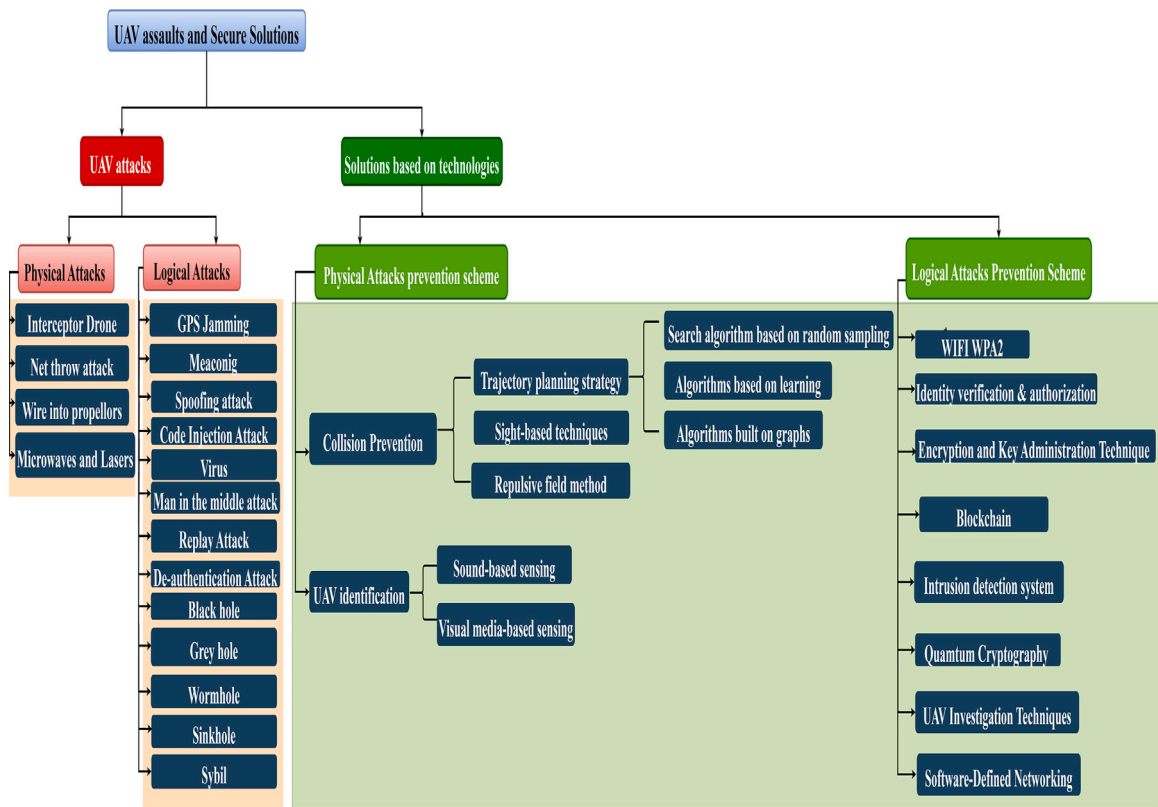


Fig. 6. Hierarchy of UAV attacks and Solutions.

In [4] the authors' main focus was on the vulnerabilities of UAVs belonging to specific domains such as military, civilian, and terrorism. Still, the authors failed to describe the physical and logical attack hierarchy and the specific frequency range and wavelength for different bands. Also, the discussed defense strategies for secure communications in IoT-based UAV networks are limited, i.e., IDS and some of the encryption schemes like traceable and privacy-preserving authentication. The author does not describe many limitations, such as lightweight authentication schemes and the latest encryption strategies for resource-constrained IoT-based UAVs. The author of this paper has not categorized the secure communication methodologies based on authentication and encryption strategies for a clear understanding, which is covered by our paper.

In [5] The author has described the usage of UAVs in civilian, military, and industry with a detailed description of UAVs and their types. The paper covers the recent scenario about UAV attacks like attacks based on terrorist activities such as Abqaiq-khuraish attack, the Caracas drone attack, and military attacks but fails to explain the taxonomy of all existing threats in IoT-based UAV networks. The paper provides the preliminary of wireless communication but has not given any frequency range and wavelength description. The recent attacks case studies were provided by the author but there was no discussion about existing state of art countermeasures and future directions provided by the authors.

In [12] The author of this paper has described the comprehensive survey on security vulnerabilities and provided a descriptive taxonomy of security threats. There is no discussion about the preliminary of UAVs, their types, and wireless communication which does not give ground understanding to the researchers and first-time readers. The author has talked about the various security vulnerabilities related to physical attacks only but has not covered logical attacks. The author has focused only on physical security and cellular communication technologies like mmWave, NOMA, massive MIMO, etc. The authors have covered some recent drone attacks but not covered war flying and Bluetooth and WIFI-related attacks. Similarly, the author has covered only the last four years' papers but this paper has covered the last six years' papers. The authors have discussed the mitigation technique for physical attacks [9] like collision i.e. trajectory planning but have not discussed any trajectory planning strategy like algorithms based on random sampling, graph, and learning which is covered in this paper. The author has not provided solutions with lightweight authentication as techniques well as quantum-based cryptography which is mentioned in this paper. The taxonomy of all threats is not picturized in one place in previous work. The AI-based Intrusion detection system & lightweight authentication schemes are presented in this paper which was not covered in previous work done by the author.

Table 2
Related work.

Reference	Year	Hierarchy	Physical attacks	Logical attacks	Security assessment	Secure communication methodology	Current research issues and Future Scope	Description
[1]	2017	×	×	∂	✓	★	×	Proposed only one methodology based on identity-based encryption
[2]	2018	×	×	∂	✓	★	×	Proposed security framework to protect an aircraft with a short review
[3]	2019	✓	×	∂	×	∂	✓	Covers few logical attacks defense strategies
[4]	2020	✓	✓	✓	×	∂	✓	Covers few prevention scheme based on IDS only
[5]	2021	✓	✓	∂	×	×	×	Describes only current perspective of security of UAVs
[12]	2022	✓	×	∂	∂	∂	✓	Covers few security threats and defense techniques
This survey	2023	✓	✓	✓	✓	✓	✓	This survey covers major security assaults and their defense strategies

✓- Mentioned ×- not mentioned ∂ - partial information ★ - limited to one.

5. Research methodology

This section covers the search strategies and their criteria to systematically carry out the study. The systematic literature review's objective is to assess, look into, and synthesize all of the existing research on this topic. To give a response to the formulated research question we follow the guidelines of Kitchenham and Charters [25]. The number of steps required to conduct a systematic literature survey are: (1) Identify the research question (2) Determine the Database (3) Locate Keyword (4) Inclusion and Exclusion Criteria (5) Search Results and research gaps. The flow chart of a systematic literature survey is shown in Fig. 7 and the survey methodology is shown in Table 3.

5.1. Identify the research questions.

In this section, the key concerns and difficulties in the IoT-based UAV Networks security sector are noted such as attacks, and vulnerabilities with recent assaults in UAV networks, communication technologies, and existing surveys. Also, the state of art defense strategies. The Research aims and questions are presented in Sections 1.2.1 and 1.2.2 respectively.

5.2. Determine the database

In this section, several internet databases, including IEEE Xplore (<https://ieeexplore.ieee.org/Xplore/home.jsp>), Springer (<https://www.springer.com>), MDPI (<https://www.mdpi.com>), Wiley Online Library (<https://onlinelibrary.wiley.com>), Engineering Village (<https://www.engineeringvillage.com>), and Google Scholar (<https://scholar.google.com>) may be used to find published academic research papers and journals.

5.3. Locate keyword

In this section, Keywords used to locate the interested paper/article are “UAV security”, “Secure communication in UAV”, “security threats and countermeasures in UAV”, “Encryption and Key management schemes”, “authentication schemes or framework in UAV”, “Blockchain in drones security”, “Intrusion Detection System Scheme in drones security”, “Quantum cryptography in

Table 3
Research methodology.

Property	Category
Research questions	• RQ1 What are the various security vulnerabilities or attacks through which communication in IoT-based UAV networks is compromised? • RQ2 What is the cutting-edge status in terms of high-end UAV security? • RQ3 What communication technologies are employed to operate UAVs? • RQ4 What makes this survey different from the existing survey? • RQ5 What are the current defense strategies available for ensuring secure communication in IoT-based UAV networks?
Databases	• IEEE Xplore: https://ieeexplore.ieee.org/Xplore/home.jsp • Springer: https://www.springer.com/ • MDPI: https://www.mdpi.com/ • Wiley Online Library: https://onlinelibrary.wiley.com/ • Engineering Village: https://www.engineeringvillage.com/ • Google Scholar: https://scholar.google.com/
Search string	• UAV security • Secure communication in UAV • security threats and countermeasures in UAV
Inclusion criteria	• Papers must be written in English vehicles • Publication Year 2017–2022 • Journal papers and conference papers • Paper based on secure UAV communication • Comprehensive survey papers including security attacks and solutions both.
Exclusion criteria	• White papers • Low-Quality Papers • Non-English Papers • Short review papers • Similar Content Papers • Paper which does not contain security attacks and countermeasures in UAV communication

secure communication of UAV”, “UAV Forensics” An example of one of the expert search queries is: (((UAV security OR Secure communication in UAV OR security threats and countermeasures in uav) WN ALL)) AND ((2022 OR 2021 OR 2020 OR 2019 OR 2018 OR 2017) WN YR)) where WN stands for “within”.

5.4. Inclusion and exclusion criteria

Papers must be written in English. Papers are limited to the duration between 2017–2022. This paper is based on security threats and their prevention schemes. This work considers journal papers, transaction papers, and conference papers based on secure UAV communication, including a comprehensive survey on security attacks and solutions, whereas this work does not include white papers, low-quality papers, Non-English papers, and articles with similar content and short review. We applied the inclusion and exclusion criteria on the title, abstract screening, and at last whole text has gone through screening.

5.5. Search results and research gaps

This section denotes the filtered six papers from the previous six years based on inclusion and exclusion criteria and the comparative analysis with gaps is presented in Table 2 based on the following parameters:

1. **Hierarchy:** Hierarchy classifies and organizes the data into different categories. Some of the filtered papers were found with the hierarchy of attacks and countermeasures. Hierarchy provides a clear view of the research topic to the researchers for better acquisition of knowledge.
2. **Physical Attacks:** The IoT-based UAV networks are vulnerable to physical attacks such as net throw attacks and Only a few papers were found with these kinds of attacks.
3. **Logical Attacks:** The IoT-based UAV networks are also vulnerable to logical attacks such as Jamming, Wormhole attacks, and Most of the papers with partial information are found for logical attacks.
4. **Security Assessment:** Security analysis is required for the proposed method or scheme whether the scheme prevents attacks or not. Few papers were found with formal and informal security analysis such as AVISPA(Automated Validation of Internet Security Protocols and Applications), ProVerif, etc.
5. **Secure Communication methodology or countermeasures:** The countermeasures were limited to one or partial information provided in the filtered papers i.e. lack of cutting edge solution.

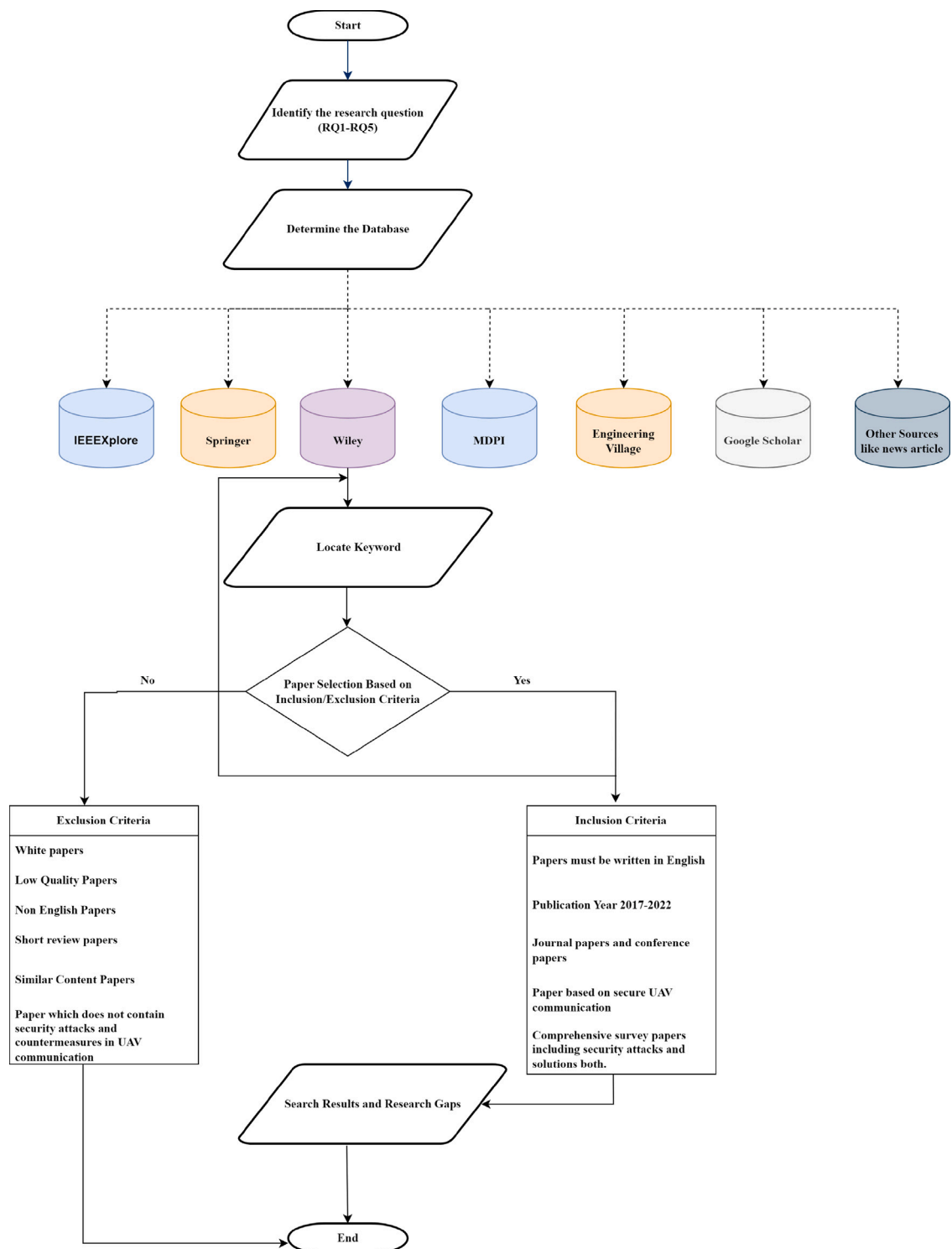


Fig. 7. Systematic literature approach.

6. **Current research issues and future scope:** Some of the papers have no information for current research issues and future directions which are essential for researchers to work upon.

6. Secure communication methodology

This section covers existing countermeasures for secure UAV communication and answers query RQ5. There are many secure communication standards for protection from attacks:

6.1. Logical attacks prevention scheme:

6.1.1. WIFI WPA2:

To defend against attacks like de-authentication, GPS spoofing, and jamming attacks, WPA2 is the best encryption technique. WPA2 is 2nd generation wireless protected access protocol that is defined as a standard in IEEE 802.11i-2004. According to this standard, a key with at least 20 characters is advised. Large-size keys are hard to crack.

6.1.2. Encryption and key administration:

As we know IoT-based UAV system is vulnerable to eavesdropping and de-authentication attacks so to mitigate these vulnerabilities a UAV system should have cryptographic and key management techniques such as:

1. **Lightweight self-certified cryptographic system:** The technique introduced is named as IoD-Crypt which is the countermeasure for authentication and confidentiality of communication [26].
2. **Identity-based encryption:** It is the better encryption technique with no overhead of certificates to use in public key architecture. It is a selection-based encipher algorithm that selectively enciphers the content and provides a data-hiding feature for confidentiality [27].
3. **Transfer key control:** The proposed approach uses cellular-based networks for drone control and provides identity verification and key control among a drone and a ground control station (GCS). A functional layer key is created between a drone and the new GCS when the drone switches from one GCS to the next. The AS key is not revealed based on the study of this method [28].
4. **OTP technology:** According to the author, in the One-time pad encryption scheme, the replicated key was provided and used with the message to obtain the cypher context using the EX-OR operation before being used with the cypher text to obtain the plain text and the replicated key to destroying the duplicated keys once the message was successfully decrypted. When compared to AES128, the performance was the best [29].
5. **Protected lightweight network coding alias:** In this author developed the client-server coding technique to combine the catchphrase with the personality which produces two keys: one for anonymization and the other for certification [30].
6. **Elliptic Curve Cryptography-Based El Gamal encryption technique:** In this technique, the optimal key is generated along with the artificial gorilla troops optimizer technique [31].

Table 4 represents encryption key and administration schemes.

6.1.3. Identity verification and authorization:

As messages are unauthenticated and their validity cannot be verified, fake signals can be injected. Deploying authentication systems that guarantee the legitimacy of only legal systems in the IoT-based UAV networks is crucial to resolving this problem.

1. **Message Authentication Code Scheme:** This type of solution uses two phases:
 - (a) **Data Verification:** The cluster head node filters using the data verification step.
 - (b) **Cluster watermark authentication:** At the sink node, this cluster verification phase is used to confirm the accuracy and integrity of the data. This scheme is vulnerable to 5 attacks i.e. propagation delay, forging of packets, data altering, replay, and gray hole attack [32].
2. **Mutual Authentication model:** The scheme is generated to solve the issue of de-authentication packets in hijacked UAVs. This scheme consists of the ground station, middleware, and UAVs. The theme is to utilize encoded channels between the above-mentioned segments [33].
3. **Continuous authentication model:** In this type of scheme, the UAV flight operators' behavior is authenticated by the flight commands sent from the UAV operator. By assuming that each operator has a distinct pattern of behavior, it is feasible to detect authorized operators who are attempting to take control of the UAVs [34].
4. **SENTINEL Framework:** The primary aim of the SENTINEL framework, a secure and efficient authentication system for unmanned aerial vehicles, is to establish mutual authenticity between UAVs and ground stations. To achieve this, the UAV must send a flight plan request to the ground station, which then decides whether to approve or reject it. If the request is approved, the ground station stores the UAV identity and flight session key in the flight information database, which can be accessed by all ground stations [35].

Table 4
Encryption and key administration techniques.

Reference	Year	Name of technique	Security assessment	Outline
[26]	2019	Lightweight Cryptographic Framework	Analysis on 2 common UAV processors,i.e. ARM(32 bit) AVR(8 bit)	- Protected from the ability to create fraudulent messages even when the attacker has access to chosen messages - Reduces energy consumption by up to 48 times compared to conventional methods.
[27]	2018	Lightweight Cryptographic Framework	Identity-based selective encryption technique	- Future and past protection of data confidentiality. - Resistance to node hijacking. - Diminish resource usage, information camouflage procedure
[28]	2017	Transfer Key control technique	Ad hoc security encryption analysis	- Single-hop key isolation. - Method ensures secure storage of communication keys.
[29]	2019	One Time pad technique	A binary additive stream cipher security analysis with C++ code	- Better security level and encryption speed. - Better accuracy than DES and AES encryption algorithm
[30]	2018	Protected lightweight network coding alias	A binary additive stream cipher security analysis with C++ code	Better security level and encryption speed.
[31]	2022	ECC-Based ElGamal encryption	Formal security analysis with a random oracle model security analysis with C++ code	Better image encryption with optimal key generation

5. **TCALAS authentication:** The proposed authentication framework for IoT-based UAV networks is based on temporal credentials and enables anonymous lightweight user authentication. It provides mutual authentication between the user and the network. The framework utilizes a three-factor authentication scheme, including cellular device, biometrics, and password.[36].
6. **Secure authentication framework:** This framework proposed was based on elliptic curve cryptosystems [37].
7. **Detectable and Privacy protecting authentication:** This technique is used for drone applications comprised of the utilization of hash function, asymmetric cryptosystems [38].
8. **Identity-based Signcryption:** The authors propose a 3-factor user access control technique-based on signcryption for IoT. This technique makes use of 3 authentication factors such as password, portable device, and biometrics [39].
9. **Privacy-preserving authentication framework:** This scheme was based on mobile edge computing. The key elements involved in this scheme are credible authority, UAVs, and, mobile edge computing devices. This framework provides an online and offline signature design with extremely efficient signature key generation and updation. This scheme helps in detecting replay attacks and repudiation threats [40].
10. **Covert channels-based lightweight authentication:** Under this scheme the author utilizes the covert channels in physical layers. The various attacks like message replay and Impersonation, eavesdropping, and MITM attack are conquered using this scheme [41].
11. **Key agreement and lightweight verification scheme:** The proposed scheme utilizes the Diffie–Hellman key exchange protocol and incorporates a symmetric key encryption algorithm to ensure message confidentiality and integrity. The authentication process between end users and UAVs is achieved through a bitwise XOR operation and a one-way hash function. Both parties mutually authenticate each other within this scheme [42].
12. **Lightweight remote user authentication with key agreement:** The author proposed this scheme when a ground user wants to gain control over the data activities from UAV directly [43].
13. **2-stage lightweight mutual authentication scheme:** The proposed scheme is designed for a multi-UAV network based on SDN. The authentication process is divided into two stages: the first stage involves the authentication between the leader drone and the ground station, while the second stage involves the authentication between the mini drone and the leader drone. The protocol, named PARTH, ensures mutual authentication, integrity, and session key security [44]
14. **Data aggregate Authentication scheme:** The proposed approach by the author suggests the utilization of ID-based encryption and elliptic curve-based technique for data aggregate authentication to achieve data security, as well as to reduce computation and communication costs. [45].
15. **Homomorphic encryption scheme:** The proposed technique by the author is called advanced linearly homomorphic authenticated encryption, which is designed to provide security against forgery and eavesdropping attacks by adversaries. Unlike other encryption techniques, LinHAE does not store the secret keys within the controller but is instead intended for use by the GCS [46].

16. **UAV-UAV and UAV-GCS security protocol based on mutual authentication:** The author proposed two secured protocols for military application [47].
 - (a) **UAV-UAV security protocol:** In this protocol secure communication among the UAVs is provided with key exchange and mutual authentication.
 - (b) **UAV-GCS security protocol:** According to this protocol, secure communication is required between UAV and the ground control station. The telemetry data and status findings are exchanged securely over the link between UAV and GCS. The protocol's security analysis was conducted using both the Scyther tool and BAN logic.
17. **Secure hash algorithm/Hash message authentication scheme:** The author has proposed the authenticated scheme based on hash message authentication. This scheme provides security against various attacks such as privileged-insider attacks, DDoS attacks, Stolen verifier attacks, replay, and spoofing attacks. The storage and calculation are less overhead under this scheme [48].
18. **3-factor authentication and key agreement protocol:** The 3-factor authentication and key agreement protocol has been proposed by the author, which utilizes the Boyko–Peinado–Venkatesan (BPV) pre-calculation and FourQ. The security of this protocol is verified through detailed analysis, ensuring the protection of forward secrecy. Furthermore, experimental findings conducted on Raspberry Pi reveal that 4Q curve-based methods are between four to five times more efficient than conventional EC curve [49].
19. **Robust Authentication key management protocol (RAMP):** The RAMP protocol is based on two types of cryptography techniques one is based on authenticated encryption primitives and the other is EC cryptography. This protocol is validated using the Scyther tool and random oracle model. This protocol provides protection against active eavesdropping and replay attack. The verification results prove that the RAMP protocol ensures security against various covert security attacks. This security protocol delivers a protected mechanism with less overhead of calculation and storage [50].
20. **2-factor lightweight verification scheme:** The author has suggested this strategy which is built on an asymmetric cryptographic technique. The previously proposed scheme was prone to lose access to IoT-based UAV networks because of corruption in key management systems. The proposed technique has proven to be secured system than other schemes. This scheme is secured against many attacks such as phishing and replay attack [51].
21. **ECC and symmetric encryption scheme:** The security of UAVs and users is ensured by utilizing Elliptic Curve Cryptography (ECC) and Symmetric Encryption in this scheme. This scheme provides anonymity and provides security against offline password guessing and various kind of security attacks like impersonation [52].

Identity verification authorization techniques are shown in Table 5.

6.1.4. Solution based on blockchain:

Blockchain technology is an emerging field in the era of the digital World. It applies to all domains apart from computer science such as energy, supply chain, and healthcare. A significant advancement in distributed ledger technology is blockchain technology. Since Satoshi Nakamoto released Bitcoin, a peer-to-peer computerized cash transaction system, its reputation has been steadily rising [53]. Blockchain technology has enormous promise in other fields where reciprocal reliance between parties is necessary. In addition to allowing safe communication between autonomous swarm systems and smart financial markets, its usefulness extends beyond electronic currency exchange systems like Bitcoin, Litecoin, etc. Blockchain provides many advantages over centralized record-keeping methods, including complete data openness and faultless operation. Blockchain offers security and privacy while doing away with the need for an intermediary or third party. Blockchain technology offers a workable and highly promising answer to the security flaws in IoT-based UAV networks. [54]. The proposed security solutions based on blockchain by different authors are the following:

1. **Neural blockchain-based Ultra reliable caching scheme:** For edge-enabled Drone networks, In [55] neural-blockchain combo was suggested. Distributed ledgers are used for trustworthy communication on the blockchain network, which can help consolidate services on the IoT-based UAV network. A hybrid neural model is suggested to keep the blockchain network's reliability criteria. The drone caching system uses the blocks to set up the operational drones in the specified network.
2. **Delivery coin-based blockchain delivery framework:** The author proposed a Delivery coin framework based on blockchain and an intrusion detection system. The blockchain uses hash functions and short signatures to achieve anonymity protection, and the identification of intrusions is done using machine learning techniques. The pBFTF protocol is a UAV-assisted forwarding technique used to achieve an accord within the blockchain-based delivery network [56].
3. **Blockchain-based Federate learning in UAVs:** The authors of [12] have examined nano UAV-edge computing for decentralization administration and safety, driven by the advantages of FL and blockchain. They have talked about basic technological structures, issues, and issues like the scalability of blockchain-assisted apps, energy economy, and transaction capacity.
4. **Key management for IoT-based UAVs using blockchain:** The authors of [57] have proposed a distributed blockchain-aided scheme for safe key management in UAV-assisted apps with a focus on the heterogeneous IoT-based UAV network. The system allows UAVs to travel between clusters on their own, spread cluster keys, and update key pairs while thwarting malicious UAVs both inside and outside the system.
5. **Blockchain-Enabled-Data Gathering Method in IoT-based UAV network:** The author [58] suggested using blockchain technology to acquire info. Blockchain technology and drone groups are proposed as data-gathering strategies to offer safety

Table 5
Identity verification and authorization.

Reference	Year	Name of technique	Security assessment	Outline
[32]	2017	Message Authentication Code Technique	Analysis based on the watermarking scheme	• Protection against data transmission delay, packet forging selective forwarding, data replay, data tampering • Minimizes the network burden and conserves power resources. • No discussion about Brute force attack.
[33]	2017	Mutual Authentication model	Security analysis by Raspberry Pi	• solve the issue of de-authentication packets in hijacked UAVs Resistance to node hijacking. • Diminish resource usage, information camouflage procedure • Prone to replay assault.
[34]	2017	Continuous authentication model	Machine learning used to train random forest classifier	• Protection against malicious commands. • Recognize authorized users. • Prone to replay assault. • Protection of privacy is not taken into account.
[35]	2020	SENTINEL Framework technique	Analysis by Proverif tool	• Offers the ability to revoke and use pseudonyms. • Vulnerable to session key attack. • No solution of geolocation confidentiality.
[36]	2019	TCALAS authentication	Analysis conducted with the help of AVISPA tool(Automated Validation of Internet Security-sensitive Protocols and Applications)	• Unrestricted modification of password or biometric data • Protection against DoS attack. • Susceptible to spoofing assault
[37]	2020	Secure authentication framework	Spontaneous security evaluation.	• Capable of withstanding sensor node takeover attack • No spoofing assaults. • No solution for geolocation confidentiality.
[38]	2020	Detectable and Privacy protecting authentication	BAN logic.	• Ensures confidentiality, integrity, and availability • Vulnerable to a session key assault.
[39]	2020	Identity-based Signcryption	Analysis done by the AVISPA tool Spontaneous security evaluation.	• Ensure prevention against sensor node capture attacks and spoofing attacks. • No solution for geolocation confidentiality.
[40]	2019	Privacy-preserving authentication framework	Security supposition.	• Identify replay assaults. • Protection against repudiation threats. • Known key attack and De synchronization attack is not discussed.
[41]	2019	ECC-Based Covert channel-based lightweight authentication	Raspberry Pi 3	• Offers adaptable networking. • Does not require traditional methods of key distribution or creation. • It does not ensure security against spoofing attacks.
[42]	2020	Key agreement and lightweight verification scheme	Ideal cipher model.	• Offers untraceable anonymity. • Protect against UAV capture attack. • No consideration of Non-repudiation.
[43]	2019	Remote user authentication with key agreement	Analysis by AVISPA tool.	• Protection against attempts to guess your password offline and unauthorized access. • No consideration for mobile edge computing devices.
[44]	2020	2-stage lightweight mutual verification scheme	Mao and Boyd Logic	• Ensures the security of both physical and session keys • Spoofing attack can be possible.
[45]	2020	Data aggregate Authentication	Ideal cipher model	• Coalition attack prevention. • Provide minimum cost and computation for UAVCN. • No protection against known key attack.

(continued on next page)

Table 5 (continued).

Reference	Year	Name of technique	Security assessment	Outline
[46]	2018	Data Homomorphic encryption technique	Ideal cipher model	• Protection against forgery and eavesdropping attacks. • No consideration of Non-Repudiation.
[47]	2021	Drone secure communication protocol (UAV-UAV and UAV-GCS security protocol based on mutual authentication)	Ideal cipher model	• Protection against forgery and eavesdropping attacks. • No consideration of Non-Repudiation.
[48]	2021	Hash message authentication technique	Ideal cipher model, ProVerif2 tool	• Protection against DoS, replay attack, stolen verifier attack, spoofing attack.
[49]	2021	3-factor authentication and key agreement protocol	Real-or-Random model.	• Secure against various known attacks like online and offline password attacks. • Attain user anonymity and untraceability. • No protection against spoofing attack.
[50]	2021	Robust Authentication key management protocol	Ideal cipher model, Scyther tool.	• Password estimation, Man in the middle, replay attack can be eliminated. • No protection against Known key attacks.
[51]	2021	2-factor lightweight verification scheme	Ad hoc security review.	• Secured against many attacks such as phishing and replay attack. • No adoption of proper security analysis technique.
[52]	2021	ECC and symmetric encryption scheme	Ideal cipher model	• provides anonymity and provides security against offline password guessing and various kind of security attacks like impersonation. • No protection against key known attacks.

as well as accuracy. To keep contact before starting data collection, the IoT devices and the autonomous aerial vehicle swarm specifically share a common key. Hash filters and electronic signatures are used to fight and thwart man-in-the-middle attacks that aim to manipulate and eavesdrop on users.

6. **Blockchain-based UAV system:** The author [59] proposed a system that uses both drones and blockchain technology for the industry. Inventory data is collected using drones, while smart contracts are enabled through the implementation of blockchain technology.
7. **Secure data propagation technique based on blockchain:** The author [60] developed a secure data distribution system for the IoT-based UAV network that utilizes game theory and blockchain technology. The system ensures the safe transmission of data by leveraging the security features of blockchain technology and the strategic decision-making principles of game theory. Blocks are verified and validated using the Proof-of-Stake (PoS) method and a forger node selection technique.
8. **Secure blockchain-based access control technique:** The author [61] outlined a secure communication system for Unmanned Aerial Vehicles (UAVs) and ground station computers, which is based on blockchain technology. This system utilizes transactions created from sensitive data gathered by GCS for block creation, which are combined with the blockchain via the ripple protocol consensus method. A Cloud-based machine communicates with GCS to facilitate this process.
9. **Blockchain-Based and ICN-Based UAS Ad Hoc Network Security** The author put forth a sophisticated and methodical approach that makes use of blockchain technology to effectively spot harmful material. This method combines interest key-content binding, on-demand authentication, and a transmitting technique. The authors created a flexible and extensible distributed consensus approach over specified network data for unmanned aerial vehicle ad hoc networks that are essential for critical operations to enable dispersed concern key-content attaching preservation to detect internal assailants [62].
10. **Smart City UAV-Based Blockchain-Based Environmental Health Tracking Program:** The authors proposed a safe method for tracking health in outdoor environments that integrates blockchain technology, Mobile Edge Computing, drones and Under this proposed system, users' devices collect their health data, which are then transmitted to a Mobile Edge Computing server by a UAV. The health information is safeguarded from cyberattacks before being sent to the MEC [63].
11. **Agent-based security inspired by blockchain:** To ensure the monitoring and security of UAV networks and to identify corrupted UAVs, The author developed a multi-agent approach built on trust rules and blockchain technology [64].
12. **Spoofing Detection in IoT-based UAVs using Blockchain:** Blockchain technology was employed by the author [65] to identify GNSS(Global Navigation Satellite System) signal assaults on IoT-based UAVs.
13. **Security infrastructure for UAV-aided wildlife monitoring based on blockchain:** The author [66] established a testbed using a public blockchain for tracking animals with the assistance of drones. The drones employ IoT devices that are linked to the animals to collect relevant data, which is transmitted to the ground control station (GCS) associated with each flying zone. The Practical Byzantine Fault Tolerance (PBFT) consensus mechanism is used by the Ground Control Station (GCS) to create and validate blocks. This consensus mechanism includes transmitting the block from the GCS to the point-to-point network of GCS nodes.

14. **Blockchain-based data delivery and gathering approach:** To ensure differentiation between drones and the ground control systems that operate them, a blockchain-based Data Delivery and Gathering system was proposed by author [67]. The DDG creates the system nodes with secret blocks and keeps track of all interactions among them.
15. **An innovative search and rescue system architecture built on blockchain:** The author [68] suggested an IoT-based blockchain architecture for performing search and rescue operations. The design uses edge computers and both small and large drones to conduct offloading tasks.
16. **Traffic management for Unmanned aerial system using blockchain:** The author [69] proposed a Lightweight blockchain-based security solution for low-altitude Drones using hyper ledger fabric that satisfies the processing and storage resource constraints of UAVs. Between the UAVs and their ground control centers, This proposed chain technique also offers secure and impermeable traffic data.

6.1.5. Solution based on intrusion detection system (IDS) scheme:

To prevent intruder attacks such as Denial of service attacks, the Intrusion detection system is required. The intrusion detection system is a monitoring system to detect any uncertain activity and make an alert for any breach by an alarm system. When an IDS is implemented in a system as complex as the FANET system, the generated alerts need to be both extremely precise and minimal to optimize the efficiency of the security system and avoid interfering with the proper operation of control applications that use the medium. The different types of security approaches carried out by IDS:

1. **Network IDS for IoT-based UAV network:** To validate the efficacy of their proposed safeguard mechanism, the author [68] utilized a paparazzi UAV, an open-source drone hardware and software project for simulated and emulated testing to accurately depict realistic scenarios. They generated three-dimensional digital signatures for denial of service (DoS) attacks by analyzing network traffic datasets that contained anomalies. The proposed technique was assessed in simple settings with a limited number of legitimate nodes and a single attack, to detect both continual and progressive flash crowds.
2. **Detection of attacks using Lightweight IDS:** The author [70] described a small, energy-efficient system that may be included in contemporary UAVs to accurately detect GPS spoofing and denial-of-service attacks. To find the attacker, the IDS combines autodidactic with a multiclass SVM(Support vector machine). The results are valuable since they were evaluated in a real-world setting with 20 UAVs, and 4 Ground Control Station.
3. **Intrusion detection and response scheme:** The author [71] developed and tried several IDSs that can handle a wide range of assaults, getting excellent precision with few false positives, to safeguard drones that conduct excursions in remote locations and to gather and transmit vital information about the conditions of these areas. The cyber-attacks like jamming, spoofing, gray hole, and black hole attacks are prevented under this scheme.
4. **IDS inspired by the human immune system:** The author [72] suggested an improved IDS that can protect drones from a variety of threats but at the expense of significant transmission overhead between nodes to find safe paths. Additionally, in highly mobile settings like Drones, the suggested IDS makes the unreal assumption that this safe path will not alter during the subsequent communication try.
5. **Secure communication with improved network IDS:** The author [73] proposed an IDS for making traffic fingerprints based on Wavelet Leader Multifractal analysis, and assessed their approach in a mixed experimental system using actual traces. Although no comparison with other techniques is provided, the method works admirably under a denial of service attack.
6. **IoT-UAV network IDS based on deep learning:** The author suggested [74] a dispersed IDS that is installed on the GCS and drones. Each UAV employs the Long short-term memory access-Recurrent neural networks algorithm to identify UAV assaults. To validate the observed attack and alert the other drones, the GCS also utilizes the Long short-term memory access-Recurrent neural networks model. On various datasets, including CICIDS2017, and TON IoT, the authors tried their model.
7. **AI-based IDS for IoT-based UAVs:** Under this scheme, the author [75] merged one-class classifiers with principal component analysis (PCA) to identify assaults so that the IDS could be trained with only regular data. Every UAV can be equipped with the suggested MAVIDS (Micro aerial vehicle IDS), which enables quick discovery and possible mitigation of cyberattacks even when there are GCS communication problems. Solution based on IDS is shown in Table 6.

6.1.6. Solution based on quantum cryptography:

As per the prior study of attacks, there are a bunch of attacks like active eavesdropping, denial of service attacks GPS jamming, and spoofing through which IoT-based UAV networks can be compromised so to overcome the possible assaults the researchers came to provide a solution based on quantum cryptography. Quantum cryptography is based on the principle of quantum physics. Quantum physics utilizes two main properties which are the following:

1. **Quantum Superposition:** Quantum superposition means combining the two valid quantum states to produce the other authorized quantum state. The quantum state can be represented in the form of a qubit in the case of quantum computers whereas traditional computer uses the concept of binary state either 0 or 1 at a particular point in time. In the case of a quantum computer, the qubit may be both states 0 or 1 simultaneously. So due to its varying state, the output of computers can never remain the same and that property is utilized by the quantum computers to solve the calculations at a high rate.
2. **Quantum entanglement:** Quantum entanglement is a property of quantum physics in which quantum particles share information, and interact with each other at some distance. This property plays an important role in quantum computing in securing communication with quantum cryptography.

Table 6
Solution based on IDS.

Reference	Year	Name of technique	Security assessment	Outline
[76]	2019	Network IDS for IoT-based UAV networks.	Attack signature with Wavelet Leader Multi-fractal	• detect both continual and progressive surge in the number of users • There is a lack of precision or exact measurements.
[70]	2019	Lightweight IDS	Analysis by anomaly based .	• Detect spoofing and Jamming attacks. • Variable selection.
[71]	2017	Intrusion detection and response technique	Signal strength intensity, JITTER	• Protection against various attacks like GPS spoofing, jamming, and false information dissemination. • Delay Tolerant network is a challenge
[72]	2020	IDS is inspired by the human immune system	Human immune system.	• Protection against wormhole, grayhole, blackhole attacks, and Fake information dissemination attacks. • High overhead of communication.
[73]	2018	Secure communication with improved network IDS	Bayesian Nash equilibrium game theory	• Protection against distributed denial of service attack. • No discussion and comparison with other techniques.
[74]	2021	IoT-UAV IDS based on deep learning	Recurrent neural networks	• Protection against well-known attacks like DDoS. • No discussion and comparison with other techniques.
[75]	2022	AI-based IDS for IoT-based UAV network	Principal component analysis	• Mitigate attacks like jamming and spoofing. • The count of UAVs is one only.

The quantum computer utilizes the above properties to provide secure communication in an IoT-based UAV network. The author [8] proposed a solution based on quantum cryptography to ensure security. The author of this paper discusses the layered architecture which consists of many layers such as a monitoring layer, UAV layer, quantum security layer (BB84 protocol), Internet layer, and control layer.

In the monitor layer, the UAVs with camera monitor and captures data from various desired positions like city, forest, etc. This layer works on mathematical expression.

In the UAV layer, UAVs appear physically in any location. To collect further information from the ground the UAV layer forms the swarm of UAVs to achieve this purpose. This layer is vulnerable to various known attacks. So, to address these problems the next quantum layer was proposed by the author.

In the Quantum layer, the data is transferred to this layer right from the monitor layer and provide security to the sensitive information using the quantum key distribution protocol namely. BB84 protocol. BB84 protocol is used for key transfer which produces a nonce private key between two entities. The BB84 protocol uses the concept of photon polarization for secure transmission.

In the Internet layer, beyond 5G mobile communication technology is used due to low latency and scalability which gives flexibility to UAV swarms for quick data transmission. 5G networks provide secure and fast communication using two communication channels such as quantum as well as the classical state.

In the last control layer, the ground control station (GCS) is responsible for centralized data control for UAVs to keep and obtain real-time data given by the above layers. The quantum key distribution between UAVs can be applied among quantum computers at GCS.

6.1.7. UAV forensics:

A new area of digital forensics called “drone forensics” seeks to gather and examine data from drone and their parts to detect assailants and malevolent intent. As UAVs travel the above-inhabited region and can be used by progressive groups and criminals to carry out unlawful activities, drone forensics is crucial. Even though forensics inquiry is well established in conventional fields, the IoT-based UAV network field lacks standards for the processes for gathering and analyzing evidence for a security event [77]. The IoT-based UAV environment presents several difficulties that prevent the standardization of forensics practices. It is challenging to develop a standard procedure that can be used with all of the different drone platforms and architectures, to start with. The investigation is also made more difficult by the drone’s multiple components and devices because forensics analysis must deal with the supporting devices as well as locate and connect the evidence gathered from these various devices. The IoT-based UAV nodes are typically connected to the cloud to offload resource-intensive duties. This feature expands the scope of the incident inquiry by moving data processing and management from the drone to the cloud. Additionally, cloud service companies rarely collaborate with

investigators, and even when they do, the data may be spread across various servers and nations. This calls for the authorization given to various organizations and agencies. The relevant methods proposed for UAV forensics are the following:

1. **Micro aerial vehicle forensic framework:** The author [78] suggested an investigative structure that was more thorough. The first division made by the authors was between hardware/physical car forensics and digital forensics for drones. Drone network data system records, sensor readings, file storage systems, and video recordings are all subject to analysis in digital forensics. Hardware forensics covers drone model identification, testing for modification, cargo carrying, fingerprint analysis, and position. The authors suggested using an investigation structure to determine the UAV's structural elements. The JAVA program was created to examine and display the drones' flying records. The only record files that can be used for this task are CSV files, though.
2. **Drone examination and analysis:** The author [79] analyzed a Parrot Bebop 2 drone to gather flying information, retrieve media from the drone, and determine ownership. They only operate with small-scale drones, though.
3. **Investigation procedure of UAV:** The efficacy of current forensic standards for forensic investigations involving UAVs and drones is evaluated by the author. The author has gone over a list of recommendations for UAV/drone inquiries. Finally, the usage of DJI Phantom 3 UAV as an intensive report to present how the suggested principles can be applied to direct a drone forensic inquiry [80].
4. **Evaluation of UAV Forensic data:** Based on the investigation that was done and the science that contributed to this work, the author suggests the UAV Death Chain and classifies the significance and difficulty of all tasks mentioned. To the best of our knowledge, no addition has evaluated UAV-related studies in cybersecurity and digital forensics using "Purple-Teaming" techniques. Additionally, this study suggests a classification structure that creates groups of UAVs with static and lives digital evidence problems according to how difficult and significant they are [81].
5. **Evaluation of UAV Forensic data:** By suggesting a digital forensic inquiry into drone technological processes, this paper develops a standardized method to carry out a digital forensic analysis of the Yuneec Typhoon H drone [82].
6. **Forensics investigation of DJI drone:** The author of this paper discusses the collection, examination, modification, and evaluation of important artifacts from the recorded flight data. To investigate and assess the relationship between the drone, the mobile phone, and the SD card, the criminal reconstruction for temporal analysis and relational artifacts is given [83].
7. **Boarded media investigation of UAV:** It is crucial to perform investigations on the drone's onboard storage media to gather forensic proof. The analysis of this storage medium, which includes various artifacts like images, movies, log files, etc., is done to compile the digital proof needed to solve a crime. This study presents the data collected from the drone storing device [84].
8. **Evaluation of UAV Forensic Analysis Software:** In [85], the author thoroughly examined the state-of-the-art UAV forensic research methods from various angles. This article also makes the following contributions:
 - (a) The discovery of personally identifiable information
 - (b) The testing and evaluation of forensic software tools currently in use.
 - (c) A review of the methods for storing data and the layout of the proof in two DJI drones (such as Phantom 4 and Matrices 210).
 - (d) The use of a three-dimensional visualization tool to examine flying paths that were retrieved from UAVs.
9. **DJI Phantom 3 UAV forensic investigation:** The author [86] provides the first comprehensive forensic examination of a DJI Phantom III drone and the first report of any proprietary file structures the drone under examination may have saved. DRone Open-source Parser (DROP), an open-source utility that analyzes private DAT files extracted from the drone's nonvolatile internal storage, is also presented. This DAT data is compressed and secured. The study also includes a preliminary analysis of written documents that were found on the mobile device that was controlling the drone and that were private, secret, and coded. These folders contained a wealth of information, including GPS coordinates, battery life, flight duration, etc. The UAV investigation scheme is presented in Table 7.

6.1.8. Solution based on SDN (Software-Defined Networking)

SDN (Software-defined Networking) is a revolutionary form of network architecture and administration that attempts to simplify and improve network operations. It adds a centralized control plane, which enables more adaptable and dynamic network administration by isolating the network's control logic from the underlying hardware architecture. Through the centralized controller's global perspective, SDN in IoT decreases the complexity of distributed IoT architectures and improves resource utilization. The different methods for SDN-based UAV networks are the following:

1. **optiML algorithm:** The authors of this paper [87] have introduced a secure machine learning-based strategy named the "optiML algorithm" to enhance the throughput of an SDN controller and improve UAV communications and security. The proposed approach comprises three main steps. Firstly, it involves the optimal positioning and user association of UAVs using a Genetic algorithm. Secondly, it determines the placement of the SDN controller using the shortest path in a single connected graph. Finally, it includes the creation and detection of DDoS attacks using a Feedforward neural network classifier.
2. **ESCALB (Efficient slave controller allocation-based load balancing) Scheme:** This work [88] offers ESCALB, a novel load balancing strategy for SDN-enabled IoT in a multi-domain scenario such as UAV networks, intending to resolve quality of service (QoS) and denial of service concerns utilizing a distributed control plan that employs SDN controllers. To do this, the ESCALB model continuously analyses the control plan's load information, allowing for the ranking of slave controllers and the efficient movement of switches within the network.

Table 7
UAV investigation schemes.

Reference	Year	Name of technique	Outline
[78]	2019	Micro aerial vehicle forensic framework	• created a Java-based program to examine and display the drones' flying records. • This framework makes use of digital and physical frameworks.
[79]	2019	Drone examination and analysis	• Obtain information related to a flight, retrieve media from UAV, and determine ownership.
[80]	2018	Investigation procedure of UAV	• Illustrate the utilization of the suggested principles in directing a drone forensic examination with the DJI Phantom 3 drone.
[81]	2021	Evaluation of UAV Forensic data	• Consider static and live digital evidence problems. • Discuss the anti-forensics techniques.
[82]	2019	Forensic validation Analysis Process	• Develops a standard method to carry out a digital forensics analysis of the Yuneec Typhoon H drone.
[83]	2019	Forensics investigation of DJI drone	• Discuss the collection, examination, modification, and evaluation of important artifacts from the recorded flight data
[96]	2022	Boarded media investigation of UAV	• Examine flying route data that has been stored.
[85]	2021	Evaluation of UAV Forensic Analysis Software	• Testing and evaluation of forensic software tools currently in use

6.2. Physical attacks prevention scheme:

6.2.1. Techniques for preventing collisions:

1. **Trajectory planning strategy:** The objective is to locate a concise and obstacle-free flight path to the intended destination, where the obstacles are stationary and well-defined. Trajectory planning methods can be further broken down into the three types of algorithms to prevent physical attacks [9] which are listed below [89]:
 - (a) **Search algorithm based on random sampling:** In this group, a collection of nodes that sample the surroundings randomly look for a route that will prevent collisions. PRM (Probabilistic Roadmap), RRBT (Rapid Exploring Random Belief Tree), RRT (Rapidly Exploring Random Tree), and DDRRT (Dynamic Domain RRT) are examples of algorithms that fall under the category of arbitrarily selecting search algorithms and have been applied to the UAV industry [90–92].
 - (b) **Algorithms based on learning** The fundamental principle of learning-based algorithms is to direct the UAV in a specific condition using a training procedure. The benefit of using learning-based approaches is being able to solve complicated, multi-objective issues. The problem of UAV route planning has been solved using some evolutionary techniques. Single and multiple UAV trajectory planning is utilized by genetic algorithms [93]. Neural networks are also used in planning a UAV trajectory such as RNN (recurrent neural network) [94]. Reinforcement learning is also used in trajectory planning of UAV [95].
 - (c) **Algorithms built on graphs:** Way planning has seen a rise in the use of graph-based search tools. With this approach, the search area is divided into a grid and the grid is represented by a collection of squares. Many algorithms are utilized for the route planning of UAVs such as Bellman-Ford, A*, Kinematic A*, Lazy theta star(θ), D* Lite. These types of algorithms are the fastest in terms of search but due to not having straightforward routes and being suitable for small areas, these algorithms are not a perfect fit [97].
2. **Sight-based techniques:** To address the impact avoidance issue, a variety of vision-based object recognition techniques have been suggested. To address the collision avoidance issue in an indoor setting, many experts used pictures taken by cameras placed on UAVs [98,99]. To estimate how near objects would be to the UAV, the researchers used a method based on stereo cameras. This approach has a high computational expense, making it unsuitable for dealing with circumstances that arise in real-time [100]. A straightforward sight-based collision avoidance strategy was suggested because it employs a monocular camera to create a collision-free route while off-board processing of heavyweight calculations [101].
3. **Repulsive field method:** The repulsive field technique is one of the effective methods used in collision prevention. It is used by many academics to address the obstacle and UAV collision avoidance issue. This method considers each UAV as a charged particle, and collision avoidance moves are produced by the repelling forces between the aircraft. The potential

field technique is a viable option for real-time application because of its straightforward execution and minimal processing complexity.[102–104]. Due to the dynamic nature of UAVs, the repulsive field method is not suitable.

6.2.2. UAV identification techniques:

The two broad categories of UAV identification techniques are the following:

1. **Sound-based sensing:** Sound sensors have been suggested in several works as an effective tool to find drones. Some studies contrast the sound signatures produced by drone engines and rotating blades with other sound signatures that have been gathered. Acoustic characteristics are taken after sound analysis. However, the precision of detection is impacted by noise and temperature. Additionally, the sensing area is constrained [84,105,106].
2. **Visual media-based sensing:** It is important to note that the majority of the research on drone identification has looked at image data and video feeds obtained from cameras [107,108]. Other literature studies have mentioned shape analysis for UAV identification [109].

UAV Investigation schemes are shown in Table 7

The hierarchy of secure communication is shown in Fig. 6

7. Current research issues and future directions

While there have been various proposed solutions aimed at enhancing the security of IoT-based UAV networks, there are still some unresolved concerns that necessitate the collaboration of researchers and industry experts.

- **Identification and avoidance of malicious software of UAV:** Research has demonstrated that malware has the potential to infect drones [110]. To the best of our knowledge, there is currently no research available that deals with the identification of malware specifically designed for drones. Despite the absence of a malware dataset specifically designed for drones, a malware classification model is recommended for ground control centers [111]. The topic of detecting malware is not new and has received extensive attention from both academic and industrial sectors. Detection of malicious software relies on operating systems and necessitates the incorporation of specific detection capabilities. To identify harmful drones, the anti-malware system must take into account certain characteristics unique to the drone's operating system, including its use of navigational state in terms of height, place, speed, and direction. Hence, a suggestion has been made for the research community to explore the subject of detecting malware in drones.
- **Unique Intrusion detection system** Many of the Intrusion detection systems will work on UAV to GCS but do not include any additional intrusions associated with the drone, GCS, or any efforts to tamper with stored data without using the cloud system. The UAV dataset is not used for IDS instead of network simulators. So, by considering the UAV dataset the IDS system can be trained.
- **Amalgamation of UAVs with different types of Networks:** To improve the secure communication between UAVs and ground stations then there is a requirement for the amalgamation of UAVs with IoT, Vehicle networks, and 5G. However, the combination of these networks can be vulnerable to attacks so an integration framework & authentication protocols should be developed by the research community to ensure secure communication.
- **Lightweight encryption algorithms:** The GCS and UAVs constantly transfer the signals in IoT-based UAV networks, resulting in a significant transmission overhead. Therefore, UAVs need a lightweight encryption protocol to protect transmission because they run on batteries and can travel great distances. The researchers can work on lightweight quantum cryptography due to the dynamic nature of UAVs.
- **Fleet system for UAV security:** In an IoT-based UAV network system, the UAVs should collaborate to complete the task but with better security, and for this purpose a fleet system is the answer to maintain secure communication among UAVs. The fleet system must include reliable node identification and collaborative learning for privacy.
- **Post Quantum cryptography:** BB84 protocol discussed in Section 6.1.6 can be vulnerable to active eavesdropping and also quantum computers can break the product of large prime numbers in the RSA algorithm using faster calculation. To protect these cons, the lattice-based cryptographic technique can be the future scope for the researchers.
- **Chaos-based cryptology:** The chaos-based cryptology technique is used for image encryption that can also be used for encrypting messages between UAVs and UAV to GCS communication. Therefore, the chances of active eavesdropping by adversaries can be eliminated by using the chaotic map.
- **Secure UAVs communication based on SDN:** Software-defined networking can help IoT-based UAV networks to eliminate security vulnerabilities with the help of controllers that can manage the whole network using the flexibility of programming rather than self-configuration. With the usage of a centralized SDN controller, the whole UAV network system can be compromised using the center point of failure. So, the researchers should come up with a solution to overcome the single point of failure.
- **Rules and guidelines:** The administration should be made to limit the unauthorized usage of UAVs for harming the population. The stringent regulations must be there by allowing only authorized people with proper licenses to operate UAVs. The government should provide a security training session to companies and UAV operators like [80].

8. Conclusion

The IoT-based UAV networks sector is gaining popularity as a promising topic that successfully assists real-time Internet of Things (IoT) applications through the integration of Unmanned Aerial Vehicles (UAVs) with IoT. The goal of IoT-based UAV networks is to extend the reach of IoT by ensuring seamless connection, scalability, dependability, and stability, as well as powerful processing and storage capabilities while minimizing energy consumption for both UAVs and IoT devices. The systematic review provided in this paper has concluded recent attacks and their possible countermeasures. There was a limited number of former research papers that consists of a hierarchy of attacks with their countermeasures based on UAV security. In this survey, we have provided an overview of UAVs, types of UAVs, communication channels, and FANET. We have provided the research methodology that consists of research questions, several databases, and inclusion and exclusion criteria to select the papers. We have done a comparison of this paper with selected papers. In this paper, we have proposed the hierarchy of attacks like physical and logical attacks, and their countermeasures. This paper has answered the research question that came across studying secure communication in IoT-based UAV networks. The state-of-the-art solution is provided in this paper with the limitation and advantages of the techniques. At last, the paper concludes the survey by providing research issues and future directions for researchers which is present in Section 7.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article

References

- [1] S. He, Q. Wu, J. Liu, W. Hu, B. Qin, Y.-N. Li, Secure communications in unmanned aerial vehicle network, in: J.K. Liu, P. Samarati (Eds.), *Information Security Practice and Experience*, Vol. 10701, in: *Lecture Notes in Computer Science*, Springer International Publishing, 2017, pp. 601–620, http://dx.doi.org/10.1007/978-3-319-72359-4_37, URL http://link.springer.com/10.1007/978-3-319-72359-4_37.
- [2] H. Sedjelmaci, S.M. Senouci, Cyber security methods for aerial vehicle networks: taxonomy, challenges and solution, *J. Supercomput.* 74 (10) (2018) 4928–4944, <http://dx.doi.org/10.1007/s11227-018-2287-8>, URL <http://link.springer.com/10.1007/s11227-018-2287-8>.
- [3] M. Riahi Manesh, N. Kaabouch, Cyber-attacks on unmanned aerial system networks: Detection, countermeasure, and future research directions, *Comput. Secur.* 85 (2019) 386–401, <http://dx.doi.org/10.1016/j.cose.2019.05.003>, URL <https://linkinghub.elsevier.com/retrieve/pii/S0167404819300963>.
- [4] J.-P. Yaacoub, H. Noura, O. Salman, A. Chehab, Security analysis of drones systems: Attacks, limitations, and recommendations, *Internet Things* 11 (2020) 100218, <http://dx.doi.org/10.1016/j.iot.2020.100218>, URL <https://linkinghub.elsevier.com/retrieve/pii/S2542660519302112>.
- [5] V. Chamola, P. Kotes, A. Agarwal, Naren, N. Gupta, M. Guizani, A comprehensive review of unmanned aerial vehicle attacks and neutralization techniques, *Ad Hoc Netw.* 111 (2021) 102324, <http://dx.doi.org/10.1016/j.adhoc.2020.102324>, URL <https://linkinghub.elsevier.com/retrieve/pii/S1570870520306788>.
- [6] D. He, S. Chan, M. Guizani, Communication security of unmanned aerial vehicles, *IEEE Wirel. Commun.* 24 (4) (2017) 134–139, <http://dx.doi.org/10.1109/MWC.2016.1600073WC>, URL <http://ieeexplore.ieee.org/document/7792372/>.
- [7] S. Tiwari, Ukraine shoots ‘laser weapons’ at Russian suicide drones, 2022, URL <https://eurasianimes.com/ukraine-shooting-laser-weapons-at-russian-suicide-drones-report/>.
- [8] V.K. Ralegankar, J. Bagul, B. Thakkar, R. Gupta, S. Tanwar, G. Sharma, I.E. Davidson, Quantum cryptography-as-a-service for secure UAV communication: Applications, challenges, and case study, *IEEE Access* 10 (2022) 1475–1492, <http://dx.doi.org/10.1109/ACCESS.2021.3138753>, URL <https://ieeexplore.ieee.org/document/9663283/>.
- [9] D. Alexander, Drone hunters: 9 of the most effective anti-drone technologies for shooting drones out of the sky, 2021, URL <https://interestingengineering.com/innovation/drone-hunters-9-of-the-most-effective-anti-drone-technologies-for-shooting-drones-out-of-the-sky>.
- [10] A. Liszewski, Parachuting quadcopter shoots off its own propellers to take out other drones, 2022, URL <https://gizmodo.com/quadcopter-shoots-down-other-drones-with-own-propellers-1848817743>.
- [11] K.-Y. Tsao, T. Girdler, V.G. Vassilakis, A survey of cyber security threats and solutions for UAV communications and flying ad-hoc networks, *Ad Hoc Netw.* 133 (2022) 102894, <http://dx.doi.org/10.1016/j.adhoc.2022.102894>, URL <https://linkinghub.elsevier.com/retrieve/pii/S1570870522000853>.
- [12] G.K. Pandey, D.S. Gurjar, H.H. Nguyen, S. Yadav, Security threats and mitigation techniques in UAV communications: A comprehensive survey, *IEEE Access* 10 (2022) 112858–112897, <http://dx.doi.org/10.1109/ACCESS.2022.3215975>, URL <https://ieeexplore.ieee.org/document/9925214/>.
- [13] P. Chiaramonte, U.S. military drones infected with mysterious computer virus, 2017, URL <https://www.foxnews.com/tech/u-s-military-drones-infected-with-mysterious-computer-virus>.
- [14] A.K. Somani, R.S. Shekhawat, A. Munda, S. Srivastava, V.K. Verma (Eds.), *Smart Systems and IoT: Innovations in Computing: Proceeding of SSIC 2019*, in: *Smart Innovation, Systems and Technologies*, vol. 141, Springer Singapore, Singapore, 2020, <http://dx.doi.org/10.1007/978-981-13-8406-6>, URL <https://link.springer.com/10.1007/978-981-13-8406-6>.
- [15] M. Faraji-Biregani, R. Fotuhi, Secure communication between UAVs using a method based on smart agents in unmanned aerial vehicles, *J. Supercomput.* 77 (5) (2021) 5076–5103, <http://dx.doi.org/10.1007/s11227-020-03462-0>, URL <http://link.springer.com/10.1007/s11227-020-03462-0>.
- [16] C. Li, Y. Xu, J. Xia, J. Zhao, Protecting secure communication under UAV smart attack with imperfect channel estimation, *IEEE Access* 6 (2018) 76395–76401, <http://dx.doi.org/10.1109/ACCESS.2018.2880979>, URL <https://ieeexplore.ieee.org/document/8533320/>.
- [17] A. Chrikhi, H. Touati, H. Snoussi, F. Kamoun, FANET: Communication, mobility models and security issues, *Comput. Netw.* 163 (2019) 106877, <http://dx.doi.org/10.1016/j.comnet.2019.106877>, URL <https://linkinghub.elsevier.com/retrieve/pii/S1389128618309034>.
- [18] L. Wang, Y. Chen, P. Wang, Z. Yan, Security threats and countermeasures of unmanned aerial vehicle communications, *IEEE Commun. Stand. Mag.* 5 (4) (2021) 41–47, <http://dx.doi.org/10.1109/MCOMSTD.0001.2000078>, URL <https://ieeexplore.ieee.org/document/9696283/>.
- [19] K. Best, J. Schmid, S. Tierney, J. Awan, N. Beyene, M. Holliday, R. Khan, K. Lee, How to Analyze the Cyber Threat from Drones: Background, Analysis Frameworks, and Analysis Tools, RAND Corporation, 2020, <http://dx.doi.org/10.7249/RR2972>, URL https://www.rand.org/pubs/research_reports/RR2972.html.

- [20] H. Alrashdeh, R.A. Shaikh, IMSI catcher detection method for cellular networks, in: 2019 2nd International Conference on Computer Applications & Information Security, ICCAIS, IEEE, Riyadh, Saudi Arabia, 2019, pp. 1–6, <http://dx.doi.org/10.1109/CAIS.2019.8769507>, URL <https://ieeexplore.ieee.org/document/8769507/>.
- [21] A. Barua, M.A. Al Alamin, M.S. Hossain, E. Hossain, Security and privacy threats for bluetooth low energy in IoT and wearable devices: A comprehensive survey, IEEE Open J. Communi. Soc. 3 (2022) 251–281, <http://dx.doi.org/10.1109/OJCOMS.2022.3149732>, URL <https://ieeexplore.ieee.org/document/9706334/>.
- [22] DGCA India, Know about UAS types, URL <https://digitalsky.dgca.gov.in/home>.
- [23] 521-2019 - IEEE Standard Letter Designations for Radar-Frequency Bands, IEEE, 2020, Place of publication not identified, OCLC: 1156377734.
- [24] S. Zaidi, M. Atiquzzaman, C.T. Calafate, Internet of flying things (IoFT): A survey, Comput. Commun. 165 (2021) 53–74, <http://dx.doi.org/10.1016/j.comcom.2020.10.023>, URL <https://linkinghub.elsevier.com/retrieve/pii/S014036642031971X>.
- [25] B. Kitchenham, Guidelines for performing systematic literature reviews in software engineering, 2007.
- [26] M.O. Ozmen, R. Behnia, A.A. Yavuz, IoD-crypt: A lightweight cryptographic framework for Internet of Drones, 2019, arXiv [arXiv:1904.06829](https://arxiv.org/abs/1904.06829)[cs].
- [27] M.S. Haque, M.U. Chowdhury, A new cyber security framework towards secure data communication for unmanned aerial vehicle (UAV), in: X. Lin, A. Ghorbani, K. Ren, S. Zhu, A. Zhang (Eds.), Security and Privacy in Communication Networks, Vol. 239, in: Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering, Springer International Publishing, Cham, 2018, pp. 113–122, http://dx.doi.org/10.1007/978-3-319-78816-6_9, URL http://link.springer.com/10.1007/978-3-319-78816-6_9.
- [28] G. Wang, K. Lim, B.-S. Lee, J.Y. Ahn, Handover key management in an LTE-based unmanned aerial vehicle control network, in: 2017 5th International Conference on Future Internet of Things and Cloud Workshops, FiCloudW, IEEE, Prague, 2017, pp. 200–205, <http://dx.doi.org/10.1109/FiCloudW.2017.81>, URL <https://ieeexplore.ieee.org/document/8113793/>.
- [29] S. Atoev, O.-J. Kwon, C.-Y. Kim, S.-H. Lee, Y.-R. Choi, K.-R. Kwon, The secure UAV communication link based on OTP encryption technique, in: 2019 Eleventh International Conference on Ubiquitous and Future Networks, ICUFN, IEEE, Zagreb, Croatia, 2019, pp. 1–3, <http://dx.doi.org/10.1109/ICUFN.2019.8806165>, URL <https://ieeexplore.ieee.org/document/8806165/>.
- [30] Y.-J. Chen, L.-C. Wang, Privacy protection for Internet of Drones: A network coding approach, IEEE Internet Things J. 6 (2) (2019) 1719–1730, <http://dx.doi.org/10.1109/JIOT.2018.2875065>, URL <https://ieeexplore.ieee.org/document/8486625/>.
- [31] F.S. Alrayes, S.S. Alotaibi, K.A. Alissa, M. Maashi, A. Alhogail, N. Alotaibi, H. Mohsen, A. Motwakel, Artificial intelligence-based secure communication and classification for drone-enabled emergency monitoring systems, Drones 6 (9) (2022) 222, <http://dx.doi.org/10.3390/drones6090222>, URL <https://www.mdpi.com/2504-446X/6/9/222>.
- [32] J. Sun, W. Wang, L. Kou, Y. Lin, L. Zhang, Q. Da, L. Chen, A data authentication scheme for UAV ad hoc network communication, J. Supercomput. 76 (6) (2020) 4041–4056, <http://dx.doi.org/10.1007/s11227-017-2179-3>, URL <http://link.springer.com/10.1007/s11227-017-2179-3>.
- [33] K. Yoon, D. Park, Y. Yim, K. Kim, S.K. Yang, M. Robinson, Security authentication system using encrypted channel on UAV network, in: 2017 First IEEE International Conference on Robotic Computing, IRC, IEEE, Taichung, Taiwan, 2017, pp. 393–398, <http://dx.doi.org/10.1109/IRC.2017.56>, URL <https://ieeexplore.ieee.org/document/7926571/>.
- [34] A. Shoufan, Continuous authentication of UAV flight command data using biometrics, in: 2017 IFIP/IEEE International Conference on Very Large Scale Integration, VLSI-SoC, IEEE, Abu Dhabi, 2017, pp. 1–6, <http://dx.doi.org/10.1109/VLSI-SoC.2017.8203494>, URL <https://ieeexplore.ieee.org/document/8203494/>.
- [35] G. Cho, J. Cho, S. Hyun, H. Kim, SENTINEL: A secure and efficient authentication framework for unmanned aerial vehicles, Appl. Sci. 10 (9) (2020) 3149, <http://dx.doi.org/10.3390/app10093149>, URL <https://www.mdpi.com/2076-3417/10/9/3149>.
- [36] J. Srinivas, A.K. Das, N. Kumar, J.J.P.C. Rodrigues, TCALAS: Temporal credential-based anonymous lightweight authentication scheme for Internet of Drones environment, IEEE Trans. Veh. Technol. 68 (7) (2019) 6903–6916, <http://dx.doi.org/10.1109/TVT.2019.2911672>, URL <https://ieeexplore.ieee.org/document/8693567/>.
- [37] Y. Kirsal Ever, A secure authentication scheme framework for mobile-sinks used in the Internet of Drones applications, Comput. Commun. 155 (2020) 143–149, <http://dx.doi.org/10.1016/j.comcom.2020.03.009>, URL <https://linkinghub.elsevier.com/retrieve/pii/S014036641930790X>.
- [38] C.-L. Chen, Y.-Y. Deng, W. Weng, C.-H. Chen, Y.-J. Chiu, C.-M. Wu, A traceable and privacy-preserving authentication for UAV communication control system, Electronics 9 (1) (2020) 62, <http://dx.doi.org/10.3390/electronics9010062>, URL <https://www.mdpi.com/2079-9292/9/1/62>.
- [39] S. Mandal, B. Bera, A.K. Sutrala, A.K. Das, K.-K.R. Choo, Y. Park, Certificateless-signcryption-based three-factor user access control scheme for IoT environment, IEEE Internet Things J. 7 (4) (2020) 3184–3197, <http://dx.doi.org/10.1109/JIOT.2020.2966242>, URL <https://ieeexplore.ieee.org/document/8957688/>.
- [40] Y. Tian, J. Yuan, H. Song, Efficient privacy-preserving authentication framework for edge-assisted Internet of Drones, J. Inf. Secur. Appl. 48 (2019) 102354, <http://dx.doi.org/10.1016/j.jisa.2019.06.010>, URL <https://linkinghub.elsevier.com/retrieve/pii/S2214212618307038>.
- [41] X.H. Cao, X. Du, E.P. Ratazzi, A light-weight authentication scheme for air force Internet of Things, in: ICC 2019 - 2019 IEEE International Conference on Communications, ICC, IEEE, Shanghai, China, 2019, pp. 1–6, <http://dx.doi.org/10.1109/ICC.2019.8761538>, URL <https://ieeexplore.ieee.org/document/8761538/>.
- [42] Y. Zhang, D. He, L. Li, B. Chen, A lightweight authentication and key agreement scheme for Internet of Drones, Comput. Commun. 154 (2020) 455–464, <http://dx.doi.org/10.1016/j.comcom.2020.02.067>, URL <https://linkinghub.elsevier.com/retrieve/pii/S0140366419319358>.
- [43] M. Wazid, A.K. Das, N. Kumar, A.V. Vasilakos, J.J.P.C. Rodrigues, Design and analysis of secure lightweight remote user authentication and key agreement scheme in Internet of Drones deployment, IEEE Internet Things J. 6 (2) (2019) 3572–3584, <http://dx.doi.org/10.1109/JIOT.2018.2888821>, URL <https://ieeexplore.ieee.org/document/8581510/>.
- [44] T. Alladi, V. Chamola, N. Kumar, PARTH: A two-stage lightweight mutual authentication protocol for UAV surveillance networks, Comput. Commun. 160 (2020) 81–90, <http://dx.doi.org/10.1016/j.comcom.2020.05.025>, URL <https://linkinghub.elsevier.com/retrieve/pii/S0140366419318456>.
- [45] W. Hong, L. Jianhua, L. Chengzhe, W. Zhe, A provably secure aggregate authentication scheme for unmanned aerial vehicle cluster networks, Peer-to-Peer Netw. Appl. 13 (1) (2020) 53–63, <http://dx.doi.org/10.1007/s12083-019-0718-9>, URL <http://link.springer.com/10.1007/s12083-019-0718-9>.
- [46] J.H. Cheon, K. Han, S.-M. Hong, H.J. Kim, J. Kim, S. Kim, H. Seo, H. Shim, Y. Song, Toward a secure drone system: Flying with real-time homomorphic authenticated encryption, IEEE Access 6 (2018) 24325–24339, <http://dx.doi.org/10.1109/ACCESS.2018.2819189>, URL <https://ieeexplore.ieee.org/document/8325268/>.
- [47] Y. Ko, J. Kim, D.G. Duguma, P.V. Astillo, I. You, G. Pau, Drone secure communication protocol for future sensitive applications in military zone, Sensors 21 (6) (2021) 2057, <http://dx.doi.org/10.3390/s21062057>, URL <https://www.mdpi.com/1424-8220/21/6/2057>.
- [48] S.U. Jan, F. Qayum, H.U. Khan, Design and analysis of lightweight authentication protocol for securing IoD, IEEE Access 9 (2021) 69287–69306, <http://dx.doi.org/10.1109/ACCESS.2021.3076692>, URL <https://ieeexplore.ieee.org/document/9418994/>.
- [49] N. Zhang, Q. Jiang, L. Li, X. Ma, J. Ma, An efficient three-factor remote user authentication protocol based on BPV-FourQ for Internet of Drones, Peer-to-Peer Netw. Appl. 14 (5) (2021) 3319–3332, <http://dx.doi.org/10.1007/s12083-021-01130-5>, URL <https://link.springer.com/10.1007/s12083-021-01130-5>.
- [50] M. Tanveer, A.U. Khan, N. Kumar, M.M. Hassan, RAMP-IoD: A robust authenticated key management protocol for the Internet of Drones, IEEE Internet Things J. 9 (2) (2022) 1339–1353, <http://dx.doi.org/10.1109/JIOT.2021.3084946>, URL <https://ieeexplore.ieee.org/document/9446966/>.

- [51] H. Khalid, S.J. Hashim, S. Mumtazah Syed Ahamed, F. Hashim, M.A. Chaudhary, Secure real-time data access using two-factor authentication scheme for the Internet of Drones, in: 2021 IEEE 19th Student Conference on Research and Development, SCOREd, IEEE, Kota Kinabalu, Malaysia, 2021, pp. 168–173, <http://dx.doi.org/10.1109/SCOREd53546.2021.9652714>, URL <https://ieeexplore.ieee.org/document/9652714/>.
- [52] S. Hussain, S.A. Chaudhry, O.A. Alomari, M.H. Alsharif, M.K. Khan, N. Kumar, Amassing the security: An ECC-based authentication scheme for Internet of Drones, IEEE Syst. J. 15 (3) (2021) 4431–4438, <http://dx.doi.org/10.1109/JSYST.2021.3057047>, URL <https://ieeexplore.ieee.org/document/9365029/>.
- [53] T. Alladi, V. Chamola, N. Sahu, M. Guizani, Applications of blockchain in unmanned aerial vehicles: A review, Veh. Commun. 23 (2020) 100249, <http://dx.doi.org/10.1016/j.vehcom.2020.100249>, URL <https://linkinghub.elsevier.com/retrieve/pii/S2214209620300206>.
- [54] P. Mehta, R. Gupta, S. Tanwar, Blockchain envisioned UAV networks: Challenges, solutions, and comparisons, Comput. Commun. 151 (2020) 518–538, <http://dx.doi.org/10.1016/j.comcom.2020.01.023>, URL <https://linkinghub.elsevier.com/retrieve/pii/S0140366419318675>.
- [55] V. Sharma, I. You, D.N.K. Jayakody, D.G. Reina, K.-K.R. Choo, Neural-blockchain-based ultrareliable caching for edge-enabled UAV networks, IEEE Trans. Ind. Inform. 15 (10) (2019) 5723–5736, <http://dx.doi.org/10.1109/TII.2019.2922039>, URL <https://ieeexplore.ieee.org/document/8734799/>.
- [56] M.A. Ferrag, L. Maglaras, DeliveryCoin: An IDS and blockchain-based delivery framework for drone-delivered services, Computers 8 (3) (2019) 58, <http://dx.doi.org/10.3390/computers8030058>, URL <https://www.mdpi.com/2073-431X/8/3/58>.
- [57] Y. Tan, J. Liu, N. Kato, Blockchain-based key management for heterogeneous flying ad hoc network, IEEE Trans. Ind. Inform. 17 (11) (2021) 7629–7638, <http://dx.doi.org/10.1109/TII.2020.3048398>, URL <https://ieeexplore.ieee.org/document/9311800/>.
- [58] A. Islam, S.Y. Shin, BUS: A blockchain-enabled data acquisition scheme with the assistance of UAV swarm in Internet of Things, IEEE Access 7 (2019) 103231–103249, <http://dx.doi.org/10.1109/ACCESS.2019.2930774>, URL <https://ieeexplore.ieee.org/document/8771158/>.
- [59] T. Fernández-Caramés, O. Blanco-Novoa, M. Suárez-Albela, P. Fraga-Lamas, A UAV and blockchain-based system for industry 4.0 inventory and traceability applications, in: 5th International Electronic Conference on Sensors and Applications, MDPI, 2018, p. 26, <http://dx.doi.org/10.3390/ecsas-5-05758>, URL <https://www.mdpi.com/2504-3900/4/1/26>.
- [60] S. Aggarwal, M. Shojafar, N. Kumar, M. Conti, A new secure data dissemination model in Internet of Drones, in: ICC 2019 - 2019 IEEE International Conference on Communications, ICC, IEEE, Shanghai, China, 2019, pp. 1–6, <http://dx.doi.org/10.1109/ICC.2019.8761372>, URL <https://ieeexplore.ieee.org/document/8761372/>.
- [61] B. Bera, D. Chattaraj, A.K. Das, Designing secure blockchain-based access control scheme in IoT-enabled Internet of Drones deployment, Comput. Commun. 153 (2020) 229–249, <http://dx.doi.org/10.1016/j.comcom.2020.02.011>, URL <https://linkinghub.elsevier.com/retrieve/pii/S0140366419318377>.
- [62] K. Lei, Q. Zhang, J. Lou, B. Bai, K. Xu, Securing ICN-based UAV ad hoc networks with blockchain, IEEE Commun. Mag. 57 (6) (2019) 26–32, <http://dx.doi.org/10.1109/MCOM.2019.1800722>, URL <https://ieeexplore.ieee.org/document/8740789/>.
- [63] A. Islam, S.Y. Shin, BHMUS: Blockchain based secure outdoor health monitoring scheme using UAV in smart city, in: 2019 7th International Conference on Information and Communication Technology, ICoICT, IEEE, Kuala Lumpur, Malaysia, 2019, pp. 1–6, <http://dx.doi.org/10.1109/ICoICT.2019.8835373>, URL <https://ieeexplore.ieee.org/document/8835373/>.
- [64] I. García-Magariño, R. Lacuesta, M. Rajarajan, J. Lloret, Security in networks of unmanned aerial vehicles for surveillance with an agent-based approach inspired by the principles of blockchain, Ad Hoc Netw. 86 (2019) 72–82, <http://dx.doi.org/10.1016/j.adhoc.2018.11.010>, URL <https://linkinghub.elsevier.com/retrieve/pii/S1570870518301689>.
- [65] R. Han, L. Bai, J. Liu, P. Chen, Blockchain-based GNSS spoofing detection for multiple UAV systems, J. Commun. Inf. Netw. 4 (2) (2019) 81–88, <http://dx.doi.org/10.23919/JCIN.2019.8917874>, URL <https://ieeexplore.ieee.org/document/8917874/>.
- [66] A. Mitra, B. Bera, A.K. Das, Design and testbed experiments of public blockchain-based security framework for IoT-enabled drone-assisted wildlife monitoring, in: IEEE INFOCOM 2021 - IEEE Conference on Computer Communications Workshops, INFOCOM WKSHPs, IEEE, Vancouver, BC, Canada, 2021, pp. 1–6, <http://dx.doi.org/10.1109/INFOCOMWKSHP551825.2021.9484468>, URL <https://ieeexplore.ieee.org/document/9484468/>.
- [67] A. Irshad, S.A. Chaudhry, A. Ghani, M. Bilal, A secure blockchain-oriented data delivery and collection scheme for 5G-enabled IoT environment, Comput. Netw. 195 (2021) 108219, <http://dx.doi.org/10.1016/j.comnet.2021.108219>, URL <https://linkinghub.elsevier.com/retrieve/pii/S1389128621002668>.
- [68] T. Nguyen, R. Katila, T.N. Gia, A novel Internet-of-Drones and blockchain-based system architecture for search and rescue, in: 2021 IEEE 18th International Conference on Mobile Ad Hoc and Smart Systems, MASS, IEEE, Denver, CO, USA, 2021, pp. 278–288, <http://dx.doi.org/10.1109/MASS52906.2021.00044>, URL <https://ieeexplore.ieee.org/document/9637730/>.
- [69] A. Allouch, O. Cheikhrouhou, A. Koubâa, K. Toumi, M. Khalgui, T. Nguyen Gia, UTM-chain: Blockchain-based secure unmanned traffic management for Internet of Drones, Sensors 21 (9) (2021) 3049, <http://dx.doi.org/10.3390/s21093049>, URL <https://www.mdpi.com/1424-8220/21/9/3049>.
- [70] M.P. Arthur, Detecting signal spoofing and jamming attacks in UAV networks using a lightweight IDS, in: 2019 International Conference on Computer, Information and Telecommunication Systems, CITs, IEEE, Beijing, China, 2019, pp. 1–5, <http://dx.doi.org/10.1109/CITS.2019.8862148>, URL <https://ieeexplore.ieee.org/document/8862148/>.
- [71] H. Sedjelmaci, S.M. Senouci, N. Ansari, A hierarchical detection and response system to enhance security against lethal cyber-attacks in UAV networks, IEEE Transactions on Systems, Man, and Cybernetics: Systems 48 (9) (2018) 1594–1606, <http://dx.doi.org/10.1109/TSMC.2017.2681698>, URL <https://ieeexplore.ieee.org/document/7890467/>.
- [72] R. Fotuhi, Securing of Unmanned Aerial Systems (UAS) against security threats using human immune system, Reliab. Eng. Syst. Saf. 193 (2020) 106675, <http://dx.doi.org/10.1016/j.res.2019.106675>, URL <https://linkinghub.elsevier.com/retrieve/pii/S0951832018314212>.
- [73] R. Zhang, J.-P. Condomines, N. Larrieu, R. Chemali, Design of a novel network intrusion detection system for drone communications, in: 2018 IEEE/AIAA 37th Digital Avionics Systems Conference, DASC, IEEE, London, 2018, pp. 1–10, <http://dx.doi.org/10.1109/DASC.2018.8569300>, URL <https://ieeexplore.ieee.org/document/8569300/>.
- [74] R.A. Ramadan, A.-H. Emara, M. Al-Sarem, M. Elhamahmy, Internet of Drones intrusion detection using deep learning, Electronics 10 (21) (2021) 2633, <http://dx.doi.org/10.3390/electronics10212633>, URL <https://www.mdpi.com/2079-9292/10/21/2633>.
- [75] J. Whelan, A. Almechadi, K. El-Khatib, Artificial intelligence for intrusion detection systems in Unmanned Aerial Vehicles, Comput. Electr. Eng. 99 (2022) 107784, <http://dx.doi.org/10.1016/j.compeleceng.2022.107784>, URL <https://linkinghub.elsevier.com/retrieve/pii/S0045790622000842>.
- [76] J.-P. Condomines, R. Zhang, N. Larrieu, Network intrusion detection system for UAV ad-hoc communication: From methodology design to real test validation, Ad Hoc Netw. 90 (2019) 101759, <http://dx.doi.org/10.1016/j.adhoc.2018.09.004>, URL <https://linkinghub.elsevier.com/retrieve/pii/S1570870518306541>.
- [77] E. Mantas, C. Patsakis, Who watches the new watchmen? The challenges for drone digital forensics investigations, Array 14 (2022) 100135, <http://dx.doi.org/10.1016/j.array.2022.100135>, URL <https://linkinghub.elsevier.com/retrieve/pii/S2590005622000066>.
- [78] A. Renduchintala, F. Jahan, R. Khanna, A.Y. Javaid, A comprehensive micro unmanned aerial vehicle (UAV/Drone) forensic framework, Digit. Investig. 30 (2019) 52–72, <http://dx.doi.org/10.1016/j.diin.2019.07.002>, URL <https://linkinghub.elsevier.com/retrieve/pii/S1742287619300799>.
- [79] F. Iqbal, B. Yankson, M.A. AlYammahi, N. AlMansoori, S.M. Qayed, B. Shah, Drone forensics: examination and analysis.
- [80] The Drone Rules, 2021, Ministry of Civil Aviation, 2022, URL <https://static.pib.gov.in/WriteReadData/specificdocs/documents/2022/jan/doc202212810701.pdf>.
- [81] F.E. Salamh, U. Karabiyik, M.K. Rogers, E.T. Matson, A comparative UAV forensic analysis: Static and live digital evidence traceability challenges, Drones 5 (2) (2021) 42, <http://dx.doi.org/10.3390/drones5020042>, URL <https://www.mdpi.com/2504-446X/5/2/42>.
- [82] F.E. Salamh, U. Karabiyik, M.K. Rogers, RPAS forensic validation analysis towards a technical investigation process: A case study of Yuneec Typhoon H, Sensors 19 (15) (2019) 3246, <http://dx.doi.org/10.3390/s19153246>, URL <https://www.mdpi.com/1424-8220/19/15/3246>.

- [83] D.-Y. Kao, M.-C. Chen, W.-Y. Wu, J.-S. Lin, C.-H. Chen, F. Tsai, Drone forensic investigation: DJI spark drone as a case study, *Procedia Comput. Sci.* 159 (2019) 1890–1899, <http://dx.doi.org/10.1016/j.procs.2019.09.361>, URL <https://linkinghub.elsevier.com/retrieve/pii/S1877050919315625>.
- [84] Z. Shi, X. Chang, C. Yang, Z. Wu, J. Wu, An acoustic-based surveillance system for amateur drones detection and localization, *IEEE Trans. Veh. Technol.* 69 (3) (2020) 2731–2739, <http://dx.doi.org/10.1109/TVT.2020.2964110>, URL <https://ieeexplore.ieee.org/document/8950116/>.
- [85] F.E. Salamh, M.M. Mirza, U. Karabiyyik, UAV forensic analysis and software tools assessment: DJI phantom 4 and matrice 210 as case studies, *Electronics* 10 (6) (2021) 733, <http://dx.doi.org/10.3390/electronics10060733>, URL <https://www.mdpi.com/2079-9292/10/6/733>.
- [86] D.R. Clark, C. Meffert, I. Baggili, F. Breitering, DROP (DRone Open source Parser) your drone: Forensic analysis of the DJI phantom III, *Digit. Investig.* 22 (2017) S3–S14, <http://dx.doi.org/10.1016/j.diin.2017.06.013>, URL <https://linkinghub.elsevier.com/retrieve/pii/S1742287617302001>.
- [87] S. Safavat, D.B. Rawat, OptiML: An enhanced ML approach towards design of SDN based UAV networks, in: *ICC 2022 - IEEE International Conference on Communications, IEEE*, 2022, pp. 1–6, <http://dx.doi.org/10.1109/ICC45855.2022.9839272>, URL <https://ieeexplore.ieee.org/document/9839272/>.
- [88] J. Ali, R.H. Jhaveri, M. Alswailim, B.-h. Roh, ESCALB: An effective slave controller allocation-based load balancing scheme for multi-domain SDN-enabled-IoT networks, *J. King Saud Univ. Comput. Inf. Sci.* 35 (6) (2023) 101566, <http://dx.doi.org/10.1016/j.jksuci.2023.101566>, URL <https://linkinghub.elsevier.com/retrieve/pii/S1319157823001209>.
- [89] Y. Zhao, Z. Zheng, Y. Liu, Survey on computational-intelligence-based UAV path planning, *Knowl.-Based Syst.* 158 (2018) 54–64, <http://dx.doi.org/10.1016/j.knsys.2018.05.033>, URL <https://linkinghub.elsevier.com/retrieve/pii/S0950705118302636>.
- [90] A. Saravanakumar, A. Kaviyarasu, R. Ashly Jasmine, Sampling based path planning algorithm for UAV collision avoidance, *Sādhanā* 46 (3) (2021) 112, <http://dx.doi.org/10.1007/s12046-021-01642-z>, URL <https://link.springer.com/10.1007/s12046-021-01642-z>.
- [91] R. DuToit, M. Holt, M. Lyle, S. Biaz, UAV Collision Avoidance Using RRT* and LOS Maximization Technical Report #CSSE12 - 03.
- [92] S. Huang, R.S.H. Teo, Computationally efficient visibility graph-based generation of 3D shortest collision-free path among polyhedral obstacles for unmanned aerial vehicles, in: *2019 International Conference on Unmanned Aircraft Systems, ICUAS, IEEE*, Atlanta, GA, USA, 2019, pp. 1218–1223, <http://dx.doi.org/10.1109/ICUAS.2019.8798322>, URL <https://ieeexplore.ieee.org/document/8798322/>.
- [93] A. Sonmez, E. Kocigit, E. Kugu, Optimal path planning for UAVs using Genetic Algorithm, in: *2015 International Conference on Unmanned Aircraft Systems, ICUAS, IEEE*, Denver, CO, USA, 2015, pp. 50–55, <http://dx.doi.org/10.1109/ICUAS.2015.7152274>, URL <http://ieeexplore.ieee.org/document/7152274/>.
- [94] K. Kelchtermans, T. Tuytelaars, How hard is it to cross the room? – Training (recurrent) neural networks to steer a UAV, 2017, arXiv [arXiv:1702.07600](https://arxiv.org/abs/1702.07600)[cs].
- [95] L. Wang, K. Wang, C. Pan, N. Aslam, Joint trajectory and passive beamforming design for intelligent reflecting surface-aided UAV communications: A deep reinforcement learning approach, *IEEE Trans. Mob. Comput.* (2022) 1–11, <http://dx.doi.org/10.1109/TMC.2022.3200998>, URL <https://ieeexplore.ieee.org/document/9869783/>.
- [96] P. Parghi, R. Dhamija, A.K. Agrawal, Innovative approach to onboard media forensic of a drone, *IOT Smart Sys.* (2022) 307–314, http://dx.doi.org/10.1007/978-981-16-3945-6_30.
- [97] M. Radmanesh, M. Kumar, P.H. Guentert, M. Sarim, Overview of path-planning and obstacle avoidance algorithms for UAVs: A comparative study, *Unmanned Syst. 06* (02) (2018) 95–118, <http://dx.doi.org/10.1142/S2301385018400022>, URL <https://www.worldscientific.com/doi/abs/10.1142/S2301385018400022>.
- [98] K. McGuire, G. de Croon, C. De Wagter, K. Tuyls, H. Kappen, Efficient optical flow and stereo vision for velocity estimation and obstacle avoidance on an autonomous pocket drone, *IEEE Robot. Autom. Lett.* 2 (2) (2017) 1070–1076, <http://dx.doi.org/10.1109/LRA.2017.2658940>, URL <http://ieeexplore.ieee.org/document/7833065/>.
- [99] P. Ramon Soria, B. Arrue, A. Ollero, Detection, location and grasping objects using a stereo sensor on UAV in outdoor environments, *Sensors* 17 (12) (2017) 103, <http://dx.doi.org/10.3390/s17010103>, URL <http://www.mdpi.com/1424-8220/17/1/103>.
- [100] A. Al-Kaff, F. García, D. Martín, A. De La Escalera, J. Armingol, Obstacle detection and avoidance system based on monocular camera and size expansion algorithm for UAVs, *Sensors* 17 (5) (2017) 1061, <http://dx.doi.org/10.3390/s17051061>, URL <http://www.mdpi.com/1424-8220/17/5/1061>.
- [101] J. Hu, H. Zhang, Z. Li, C. Zhao, Z. Xu, Q. Pan, Object traversing by monocular UAV in outdoor environment, *Asian J. Control* 23 (6) (2021) 2766–2775, <http://dx.doi.org/10.1002/asjc.2415>, URL <https://onlinelibrary.wiley.com/doi/10.1002/asjc.2415>.
- [102] H.V. Abeywickrama, B.A. Jayawickrama, Y. He, E. Dutkiewicz, Potential field based inter-UAV collision avoidance using virtual target relocation, in: *2018 IEEE 87th Vehicular Technology Conference, VTC Spring, IEEE*, Porto, 2018, pp. 1–5, <http://dx.doi.org/10.1109/VTCSpring.2018.8417773>, URL <https://ieeexplore.ieee.org/document/8417773/>.
- [103] P.K. Selvam, G. Raja, V. Rajagopal, K. Dev, S. Knorr, Collision-free path planning for UAVs using efficient artificial potential field algorithm, in: *2021 IEEE 93rd Vehicular Technology Conference, VTC2021-Spring, IEEE*, Helsinki, Finland, 2021, pp. 1–5, <http://dx.doi.org/10.1109/VTC2021-Spring51267.2021.9448937>, URL <https://ieeexplore.ieee.org/document/9448937/>.
- [104] J. Sun, J. Tang, S. Lao, Collision avoidance for cooperative UAVs with optimized artificial potential field algorithm, *IEEE Access* 5 (2017) 18382–18390, <http://dx.doi.org/10.1109/ACCESS.2017.2746752>, URL <http://ieeexplore.ieee.org/document/8022685/>.
- [105] M.Z. Anwar, Z. Kaleem, A. Jamalipour, Machine learning inspired sound-based amateur drone detection for public safety applications, *IEEE Trans. Veh. Technol.* 68 (3) (2019) 2526–2534, <http://dx.doi.org/10.1109/TVT.2019.2893615>, URL <https://ieeexplore.ieee.org/document/8616877/>.
- [106] X. Chang, C. Yang, J. Wu, X. Shi, Z. Shi, A surveillance system for drone localization and tracking using acoustic arrays, in: *2018 IEEE 10th Sensor Array and Multichannel Signal Processing Workshop, SAM, IEEE*, Sheffield, 2018, pp. 573–577, <http://dx.doi.org/10.1109/SAM.2018.8448409>, URL <https://ieeexplore.ieee.org/document/8448409/>.
- [107] H. Liu, F. Qu, Y. Liu, W. Zhao, Y. Chen, A drone detection with aircraft classification based on a camera array, *IOP Conference Series: Materials Science and Engineering* 322 (2018) 052005, <http://dx.doi.org/10.1088/1757-899X/322/5/052005>, URL <https://iopscience.iop.org/article/10.1088/1757-899X/322/5/052005>.
- [108] B. Taha, A. Shoufan, Machine learning-based drone detection and classification: State-of-the-art in research, *IEEE Access* 7 (2019) 138669–138682, <http://dx.doi.org/10.1109/ACCESS.2019.2942944>, URL <https://ieeexplore.ieee.org/document/8846214/>.
- [109] E. Unlu, E. Zenou, N. Riviere, Using shape descriptors for UAV detection, *Electron. Imaging* 30 (9) (2018) 128–128–5, <http://dx.doi.org/10.2352/ISSN.2470-1173.2018.09.SRV-128>, URL <https://library.imaging.org/ei/articles/30/9/art00005>.
- [110] S. Khandelwal, MalDrone — First ever backdoor malware for drones, 2015, URL <https://thehackernews.com/2015/01/MalDrone-backdoor-drone-malware.html>.
- [111] Y. Sung, S. Jang, Y.-S. Jeong, J.H.J.J. Park, Malware classification algorithm using advanced Word2vec-based Bi-LSTM for ground control stations, *Comput. Commun.* 153 (2020) 342–348, <http://dx.doi.org/10.1016/j.comcom.2020.02.005>, URL <https://linkinghub.elsevier.com/retrieve/pii/S0140366419319656>.